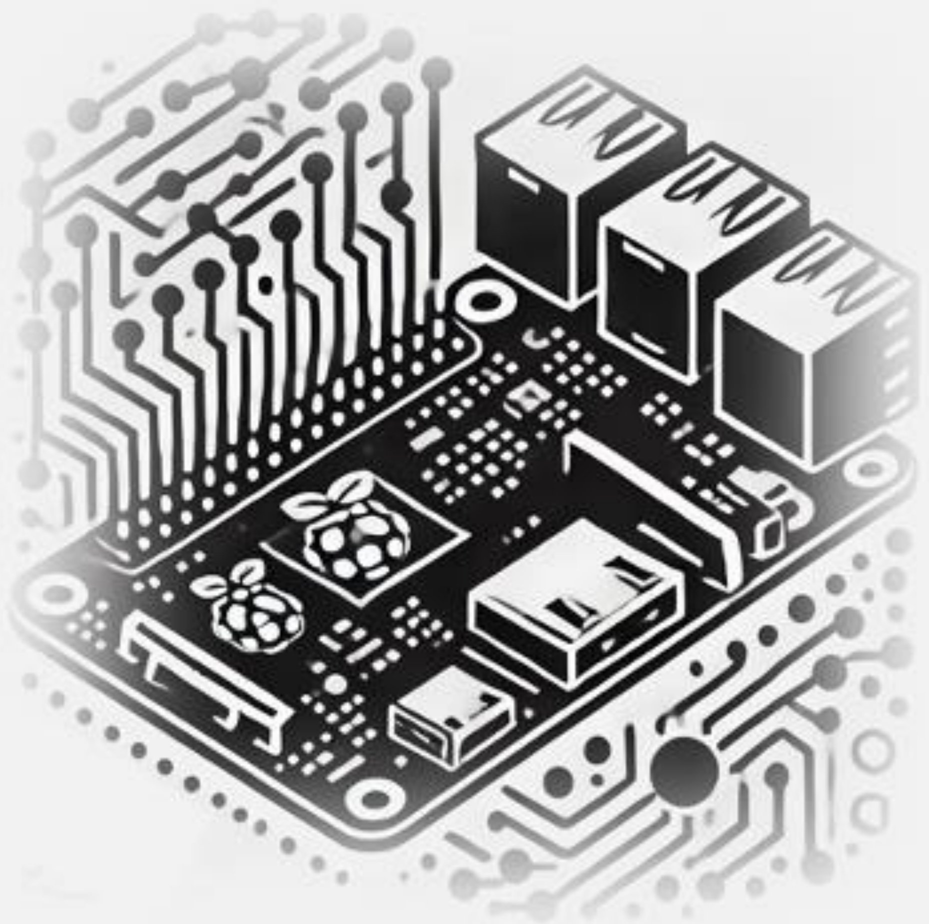
IESTI05 – Edge AI

Machine Learning System Engineering

1. Introduction to Edge AI
   About the Course & Syllabus

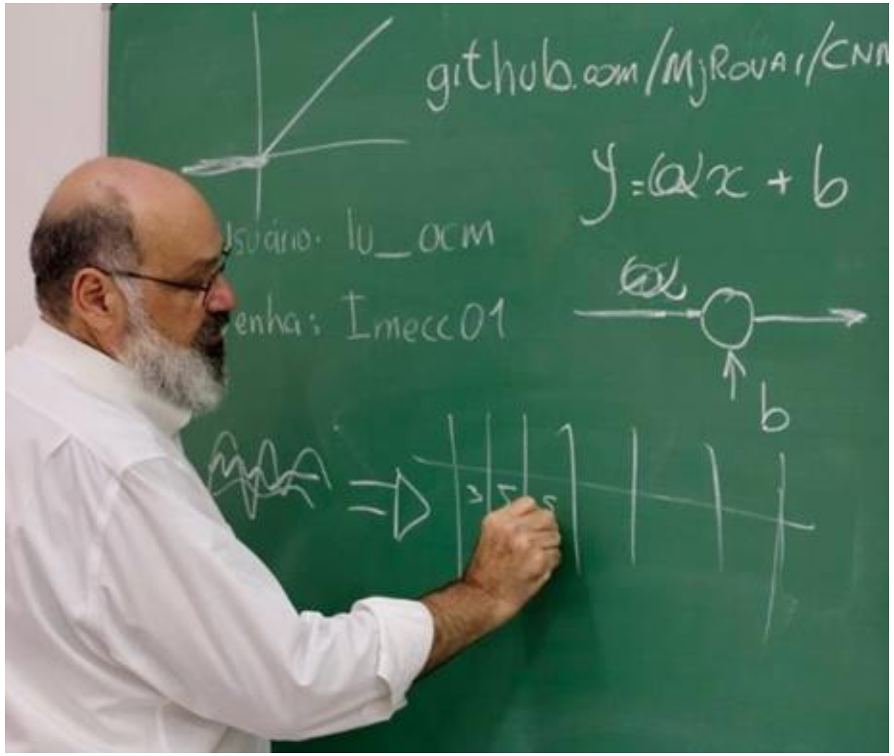


UNIFEI

Marcelo Rovai
Professor Honoris Causa

Who I am

- Brazilian from São Paulo, **Data Science Master's degree from UDD,** Chile, and **MBA from IBMEC (INSPER),** Brazil.
- Graduated in 1982 as an **Engineer from UNIFEI** with **Specialization from Poli/USP**, both in Brazil.
- Worked as a **teacher, engineer, and executive** in several technology companies such as CDT/ETEP, AVIBRAS Aeroespacial, SID Informática, ATT-GIS, NCR, DELL, COMPAQ (HP), and more recently at IGT as a Regional VP, and a Senior Advisor for Latin America.
- **Write about electronics**, publishing in sites as MJRoBot.org (Editor/Writer), Hackster.io (#1 Contributor), Instructables.com, and Medium.com (TDS – Towards Data Science).
- **Professor** at UNIFEI (IESTI), teaching "Machine Learning applied to Embedded Devices" course (IESTI01) and EdgeAI Engineering (IESTI05).
- **AiEng4D group** and the **EDGE AIP** – the Academia-Industry Partnership of **EDGEAI Foundation** Co-Chair, promoting EdgeAI education in developing countries.

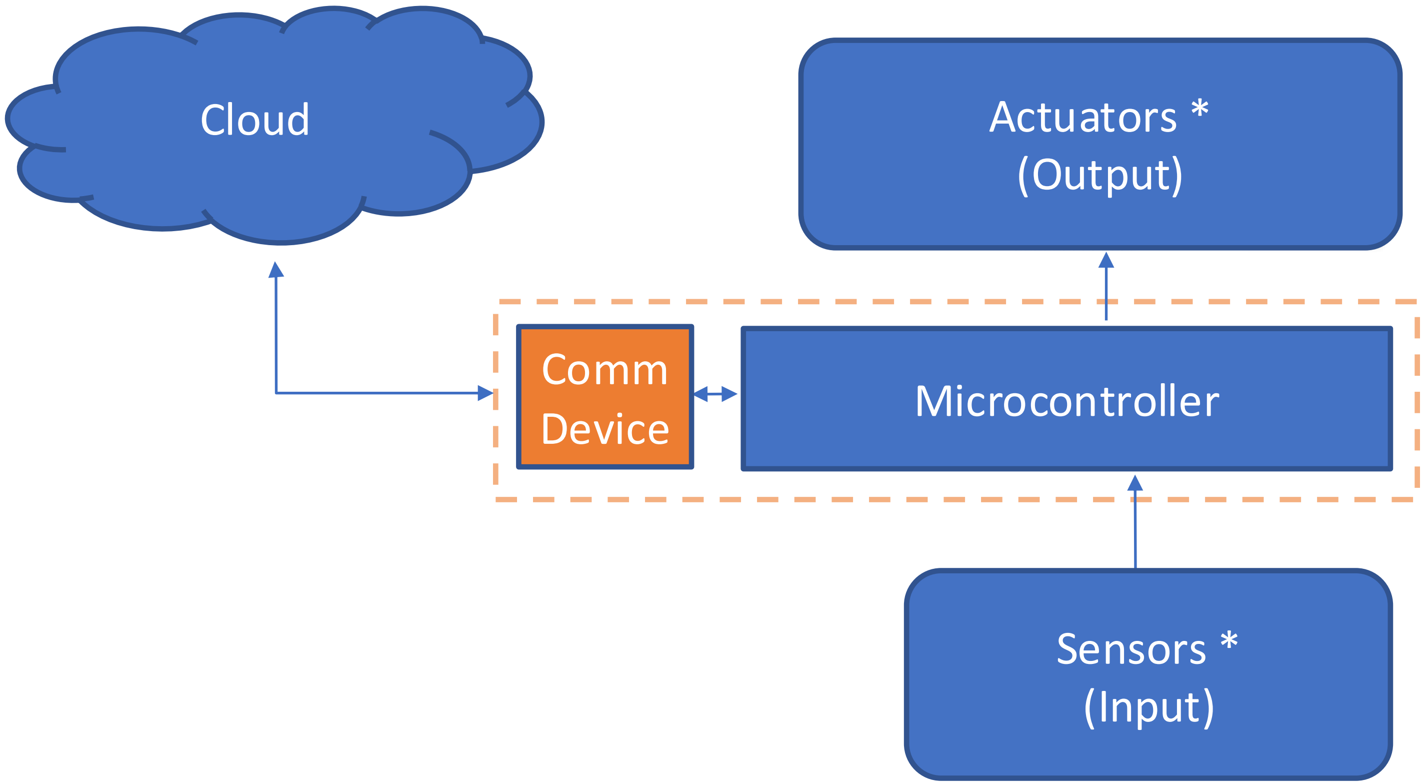


Marcelo Rovai

AIEng4D AI Engineering for Global Impact

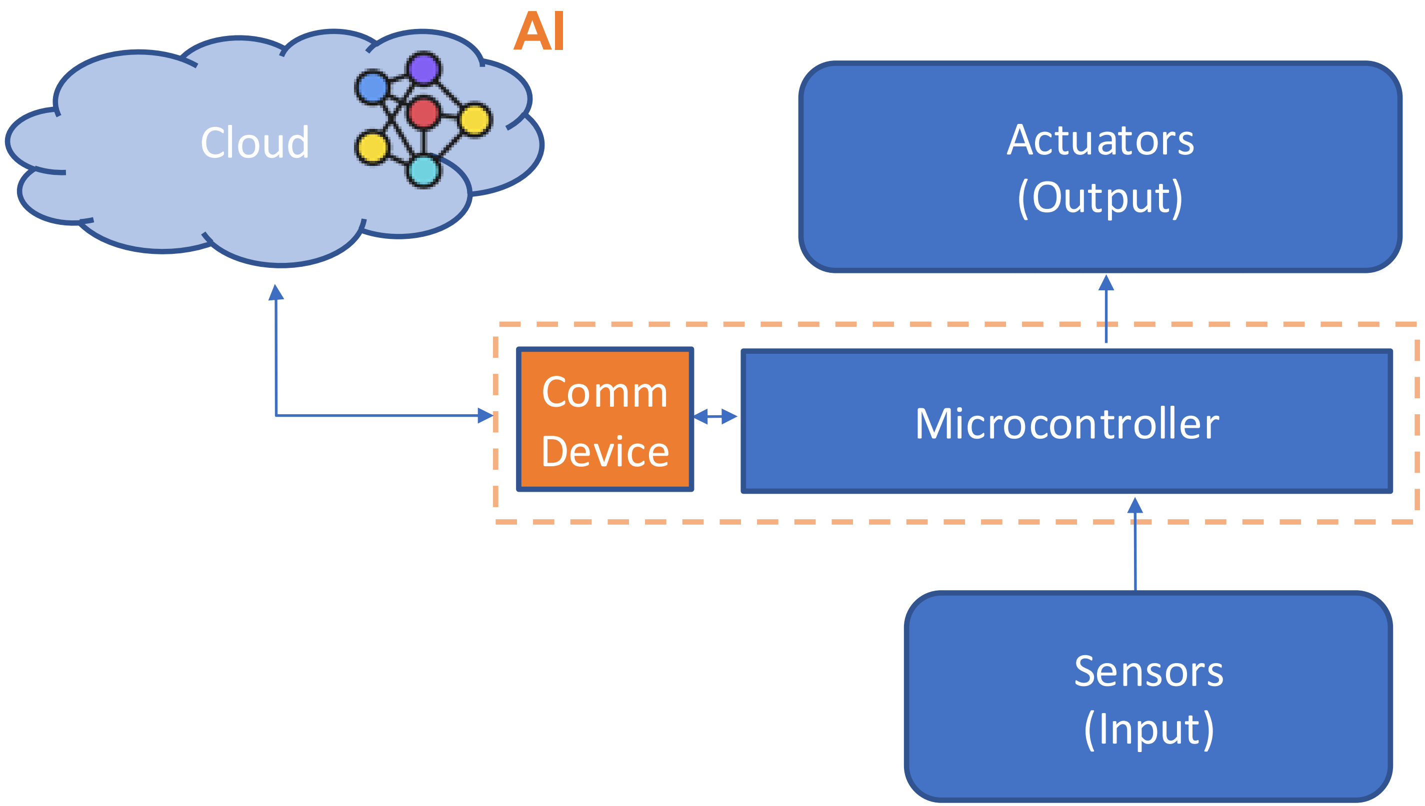
Internet of Things (IoT)

Typical IoT Project



*** "Things"**

Typical AIoT Project … … Issues

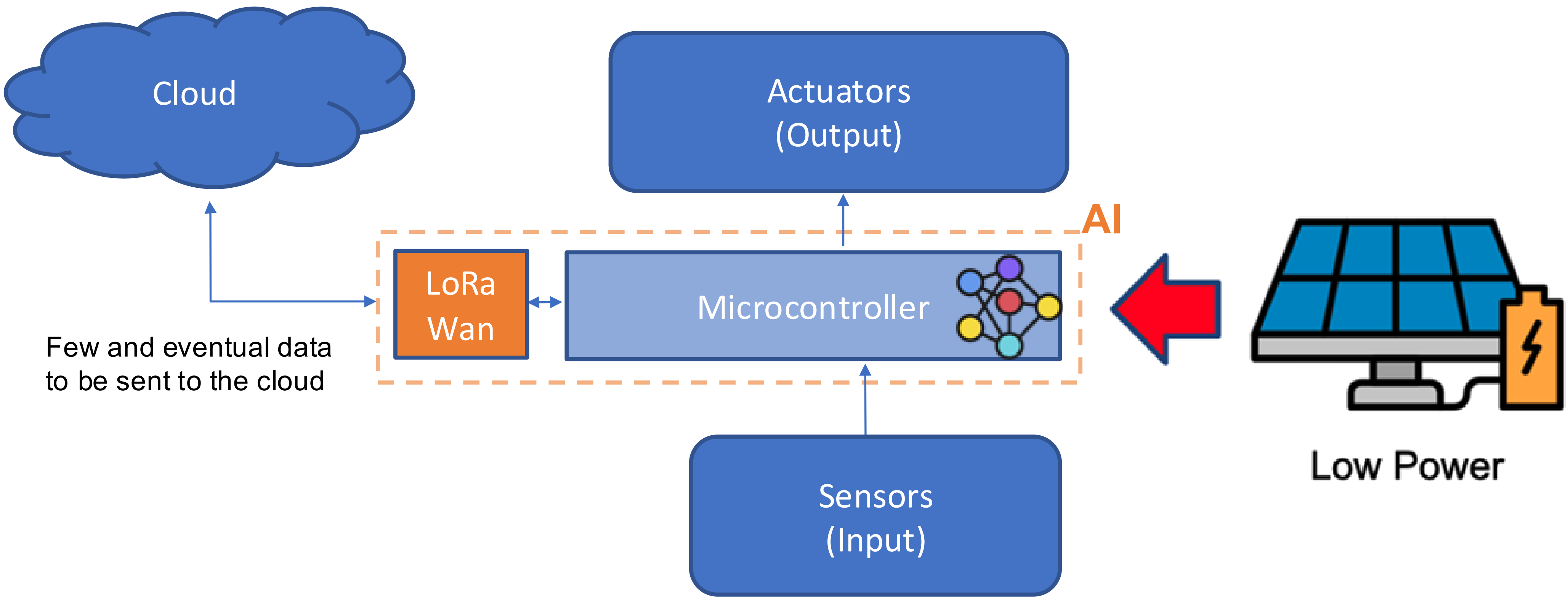


Bandwidth
Latency
Energy
Reliability
Privacy

… Solution ?

IoT 2.0 * – Edge AI/ML

* Intelligence of Things

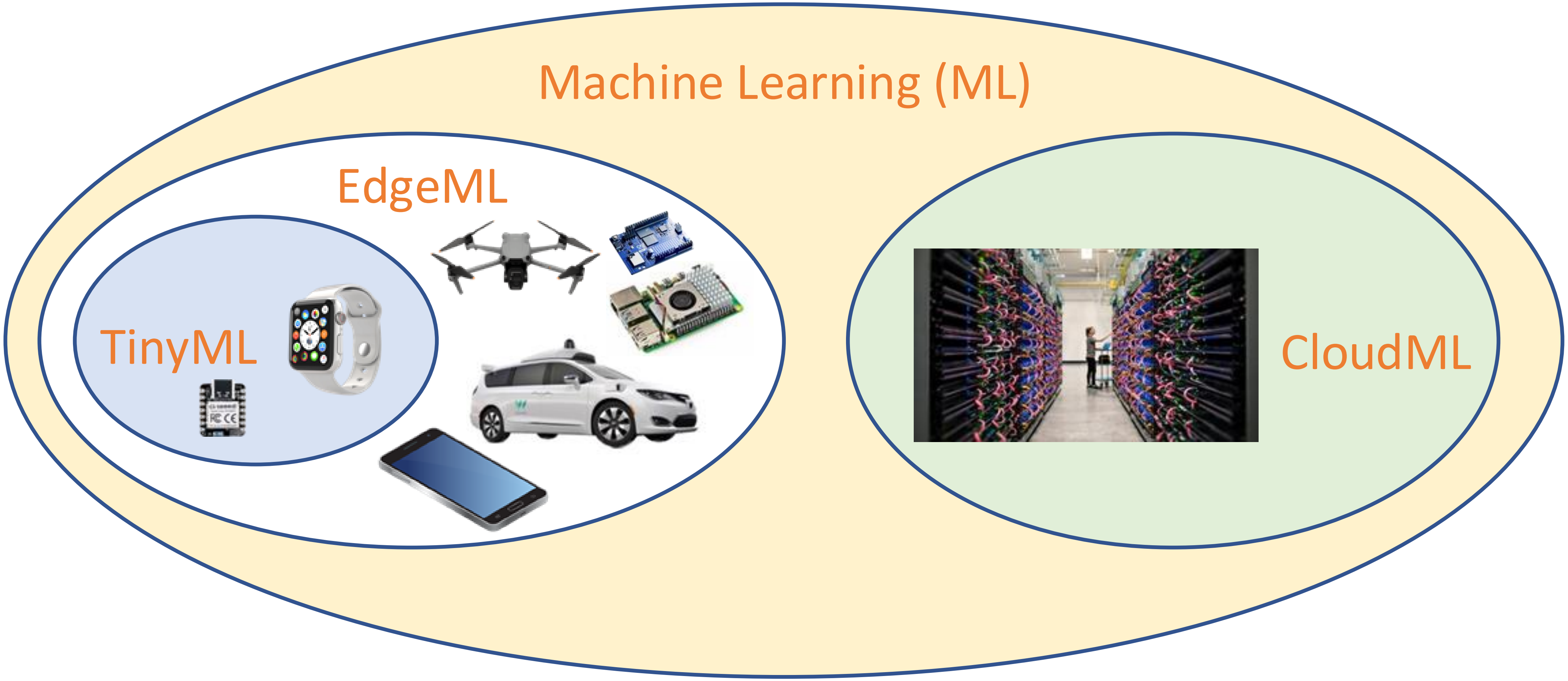


... **Solution ->** ML goes close to data

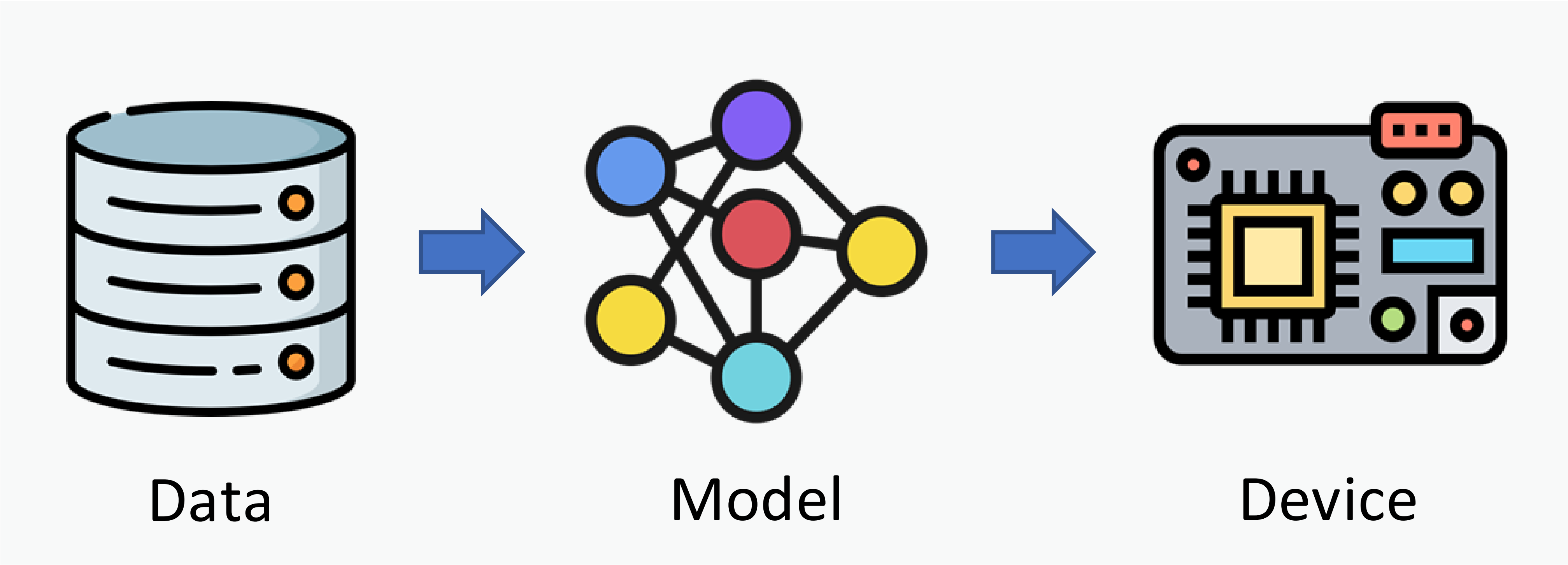
Embedded ML

EdgeAI & TinyML

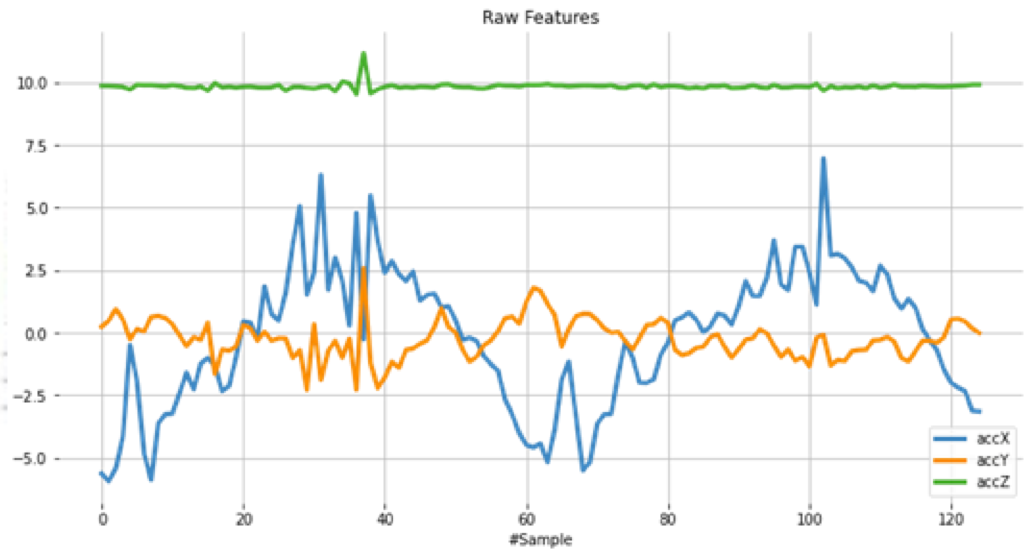
From Tiny to Cloud: One AI, Many Scales

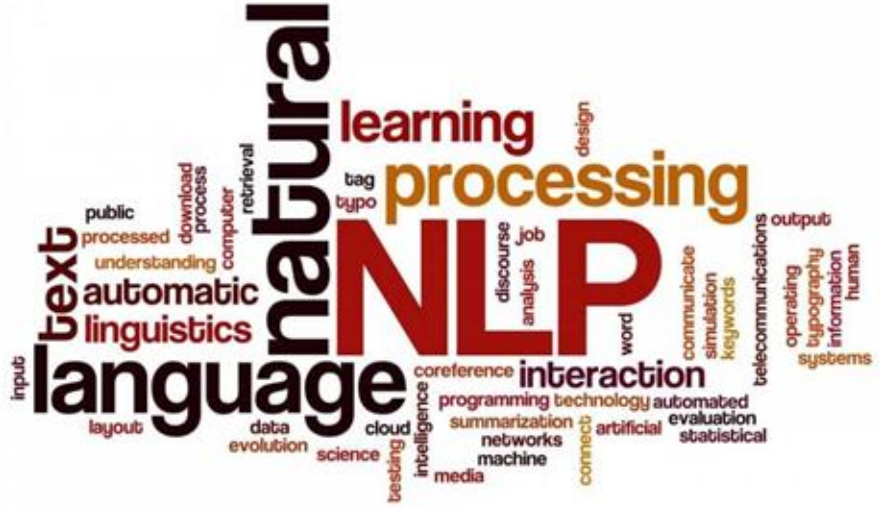


ML Deployment Pipeline

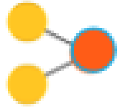


Unstructured Data





Perceptron (P)



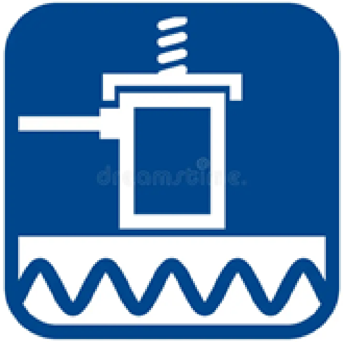
Vibration Analysis

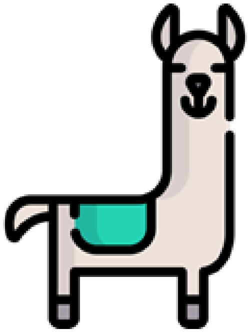
Image Classification

Text Generation

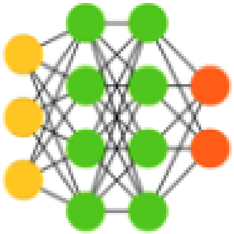
Image Generation

Large Language Models- LLMs

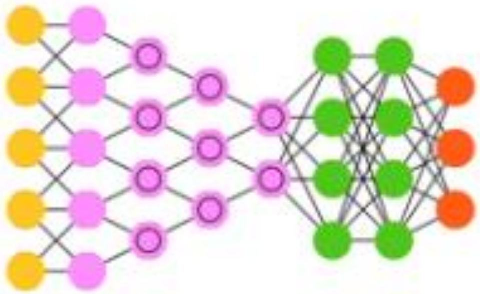




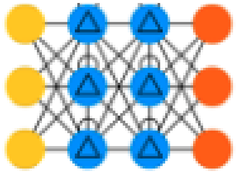
MLP - Deep Neural Network



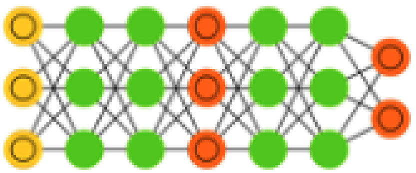
CNN - Convolutional NN



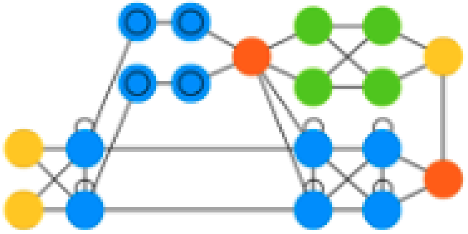
RNN - Recurrent NN (GRU/LSTM)



GAN - Generative Adversarial Net.



TF - Transformer (Attention)



Datasets Preprocessing

Quantization Pruning, Distillation

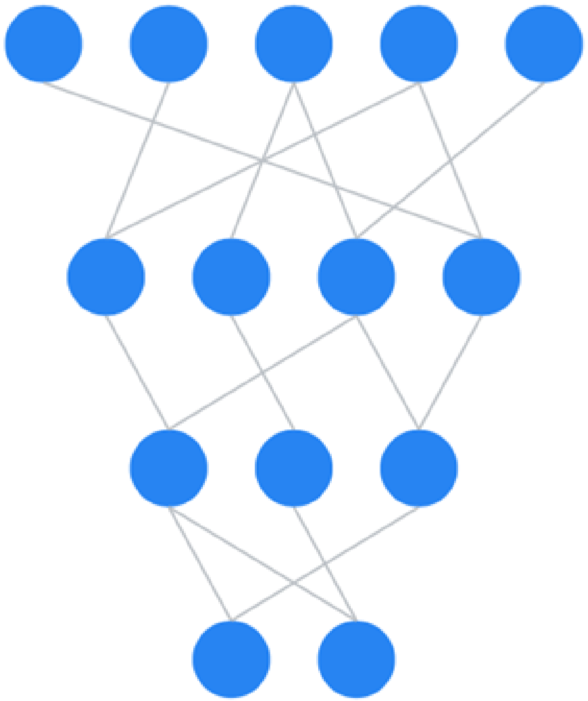
Resource constraints

Sound

Vision

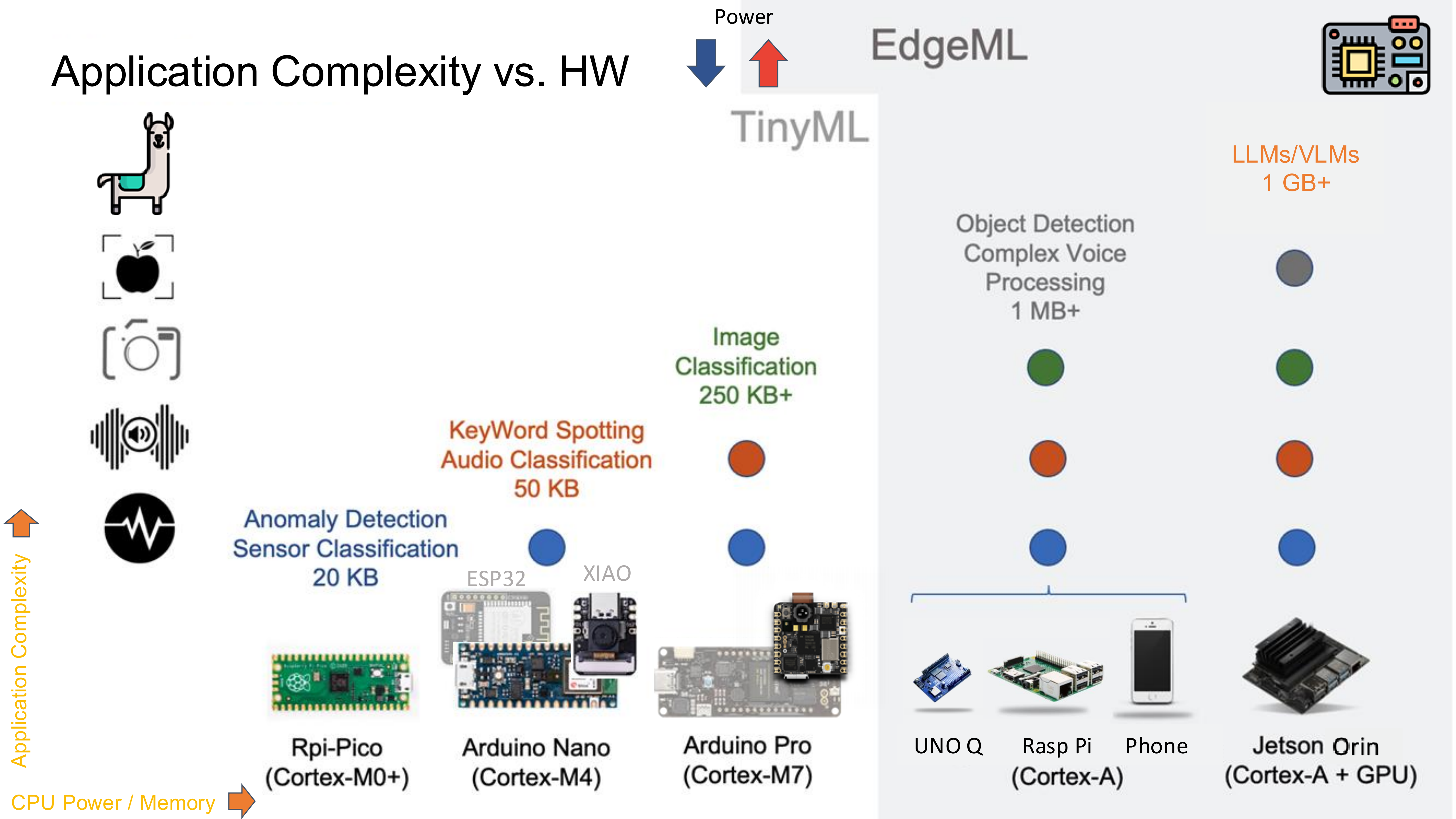
Vibration

Text

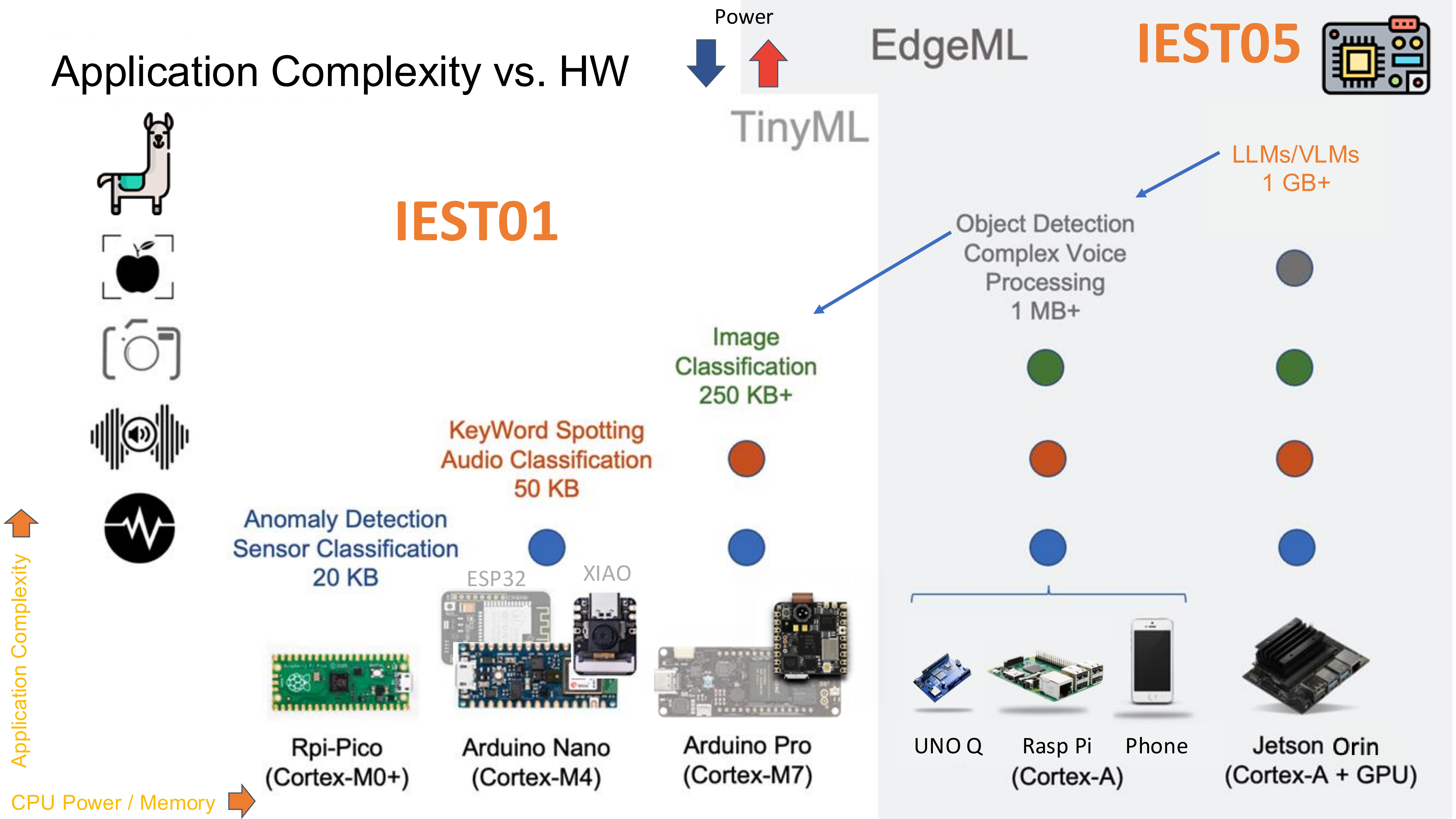


End-to-end TinyML/EdgeAI Application

Application Complexity vs. HW



Application Complexity vs. HW



Generative AI at the Edge

Generative AI (GenAI)

Generative AI is an artificial intelligence system capable of creating new, original content across various media — text, images, audio, and video. These systems learn patterns from existing data and use that knowledge to generate novel outputs that didn't previously exist.

When used for generative tasks, these model families are all types of GenAI:

LLM	SLM	VLM
Large Language Models	**Small Language Models**	**Visual-Language Models**
GPT, Claude, Gemini, Qwen	Gemma, Phi, Llama, Qwen	Gemini, Gemma, MoonDream, GPT, PaliGemma, Florence

GenAI at the Edge

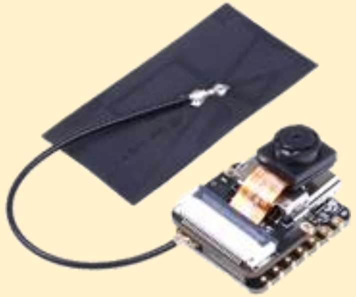
"GenAI at the Edge means the model lives on the device — not in a data center."

While LLMs typically require cloud infrastructure, compact SLMs* and small VLMs can run directly on edge devices.

* Current trends point to 2026 as the year of SLMs (Models < 7B to 10B), with specialized models surpassing generalist ones in specific tasks, such as programming and medical diagnoses.

Sustainability Matters

0.5W



5W-

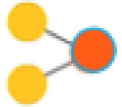
100W+

15W

20W+

100x more energy efficient than cloud AI

Perceptron (P)



Vibration Analysis

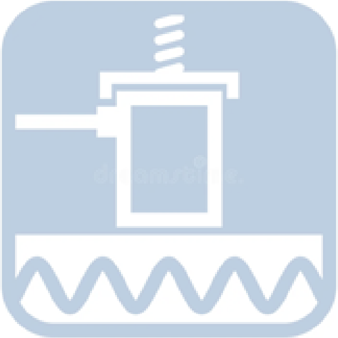
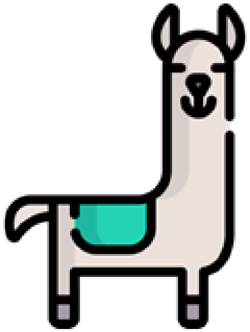


Image Classification

Text Generation

Image Generation

Large Language Models- LLMs



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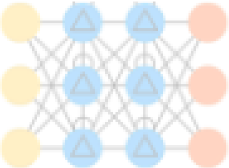

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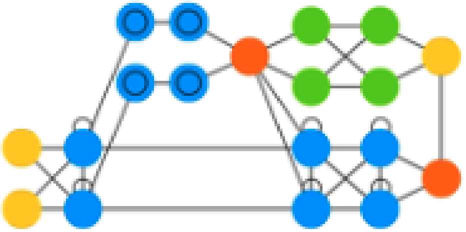


Transformers to LLMs and SLMs

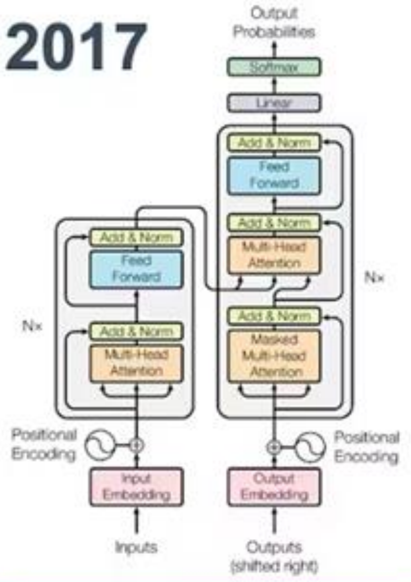
Open - China: KIMI | Qwen | DeepSeek

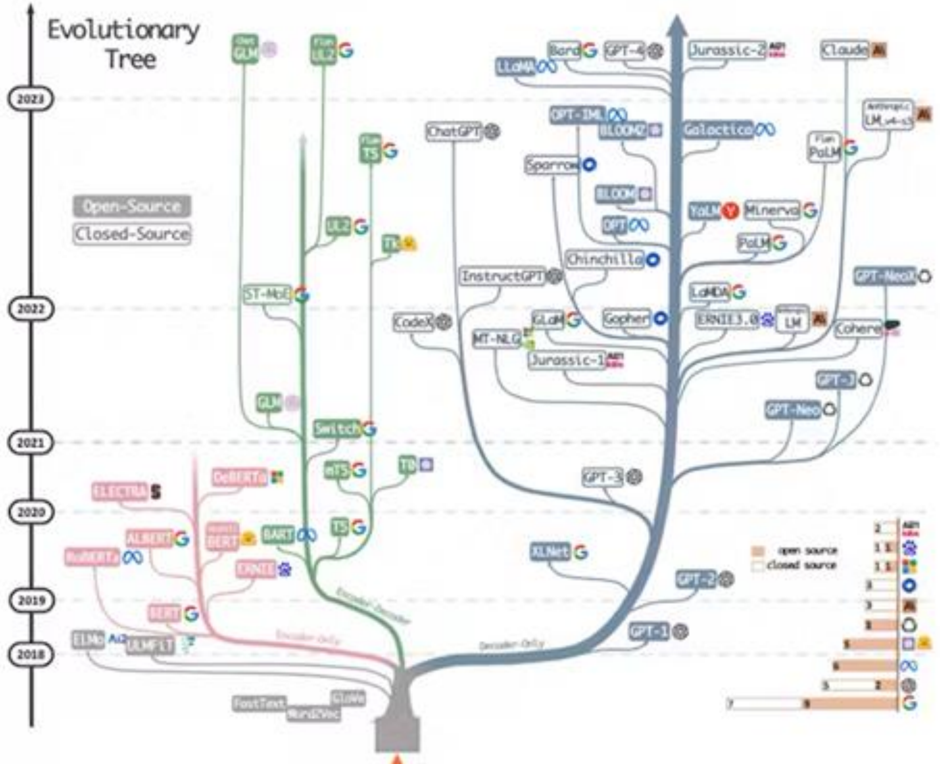
Open: Phi | Gemma | LlaMa

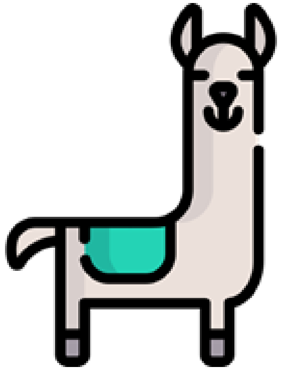
Closed: Gemini | GPT | Claude

2026: Grok

2017







LLM TRAINING

Pretraining

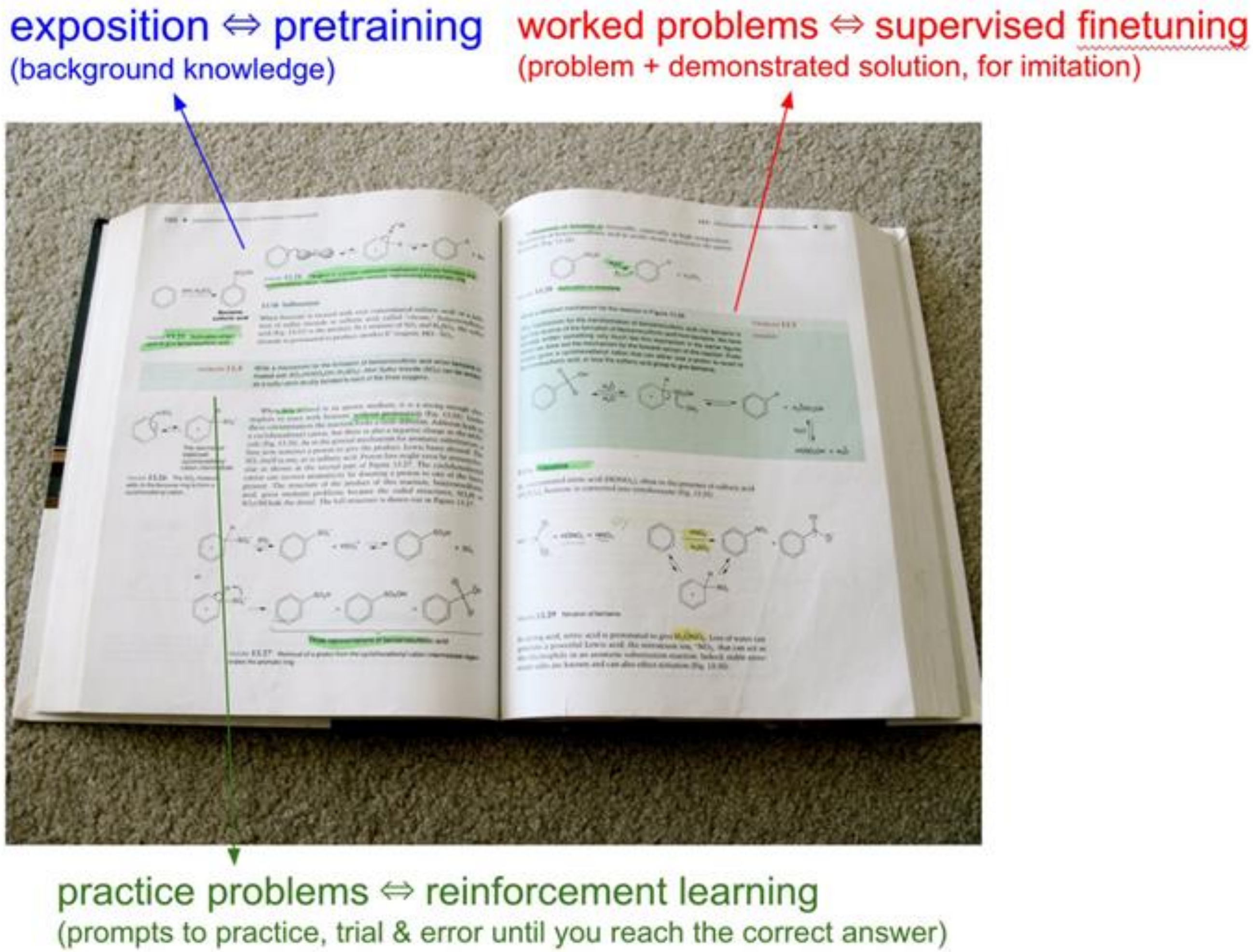
Base model — **GPT**

Post-Training
Supervised
Fine Tuning

SFT model

Post-Training
Reinforcement
Learning

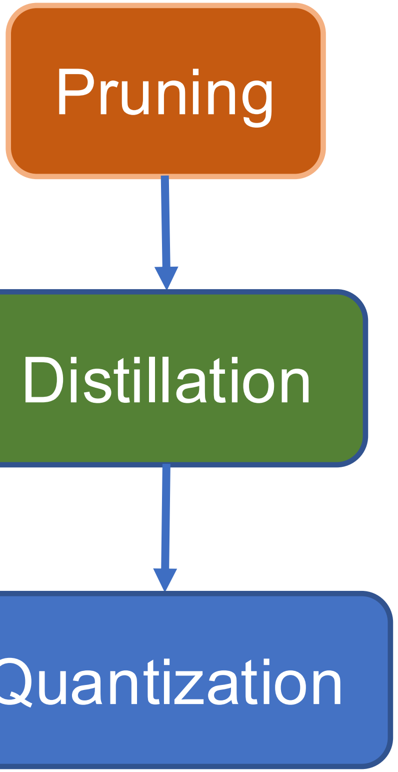
RL model — **ChatGPT**



[Deep Dive into LLMs like ChatGPT]

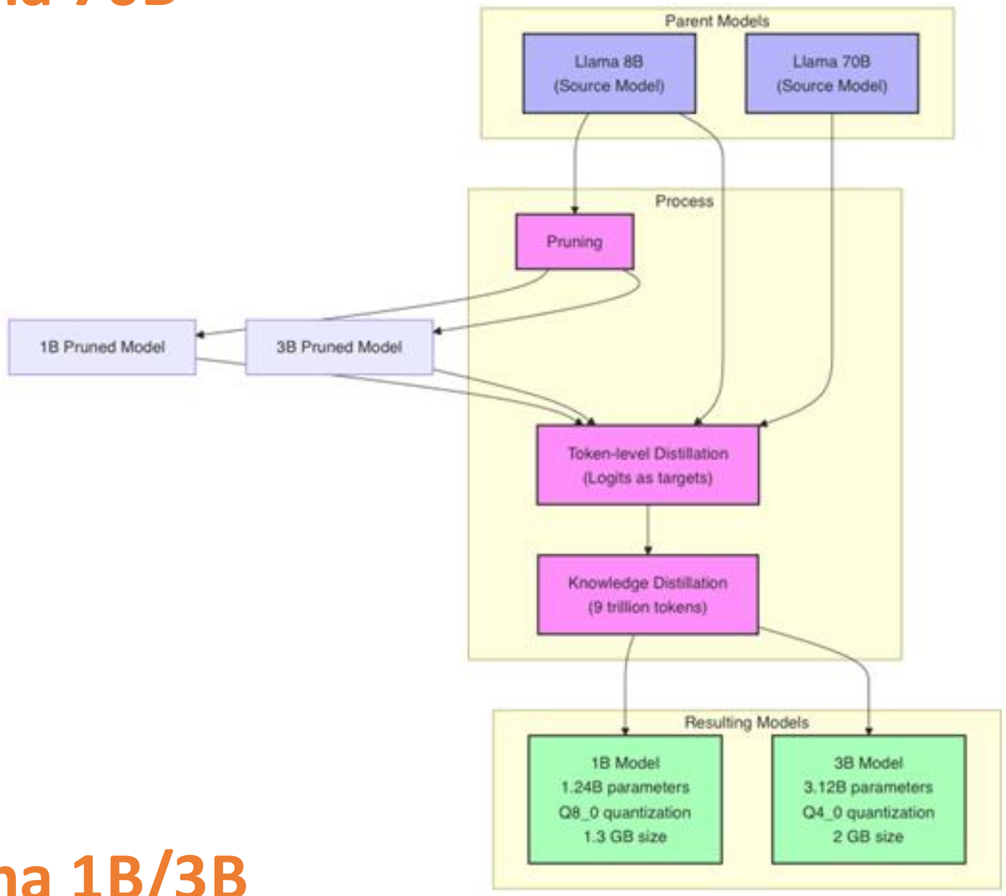
Small Language Models (SLMs)

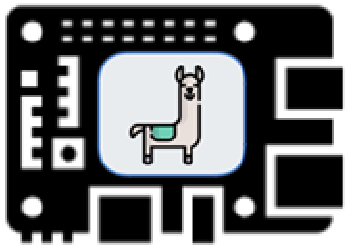
LLM **Llama 70B**



SLM **Llama 1B/3B**

Llama 1B/3B





Running SLMs locally

How to Run SLMs at the Edge

Ollama

Open-source framework for running SLMs locally. The easiest way to get started on Raspberry Pi — supports Qwen, Phi, Gemma, Llama, LLaVa, and more. Does not run easily on Arduino UNO-Q.

llama.cpp

Lightweight C++ inference engine for quantized models. Tested successfully on Arduino UNO-Q — preferred for more control, compact size, and low memory usage. Also efficient on Raspberry Pi.

HF Transformers

Python library with broad model and task coverage via the Hugging Face ecosystem. Well-supported for running SLMs on Raspberry Pi.

LiteRT-LM

Google AI Edge solution designed for production deployments. Optimized for running language models efficiently across edge devices.

Small Models

LlaMa

Gemma

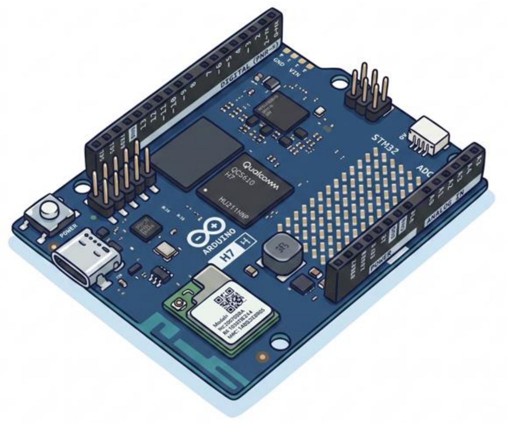
marcelo_rovai — mjrovai@raspi-5: ~ — ssh mjrovai@192.168.4.209 — 74×12

```
(ollama) mjrovai@raspi-5:~ $ ollama run llama3.2:3b --verbose
>>> What is the capital of France?
The capital of France is Paris.

total duration:       1.808927736s
load duration:        39.854862ms
prompt eval count:    32 token(s)
prompt eval duration: 221.506ms
prompt eval rate:     144.47 tokens/s
eval count:           8 token(s)
eval duration:        1.506376s
eval rate:            5.31 tokens/s
```

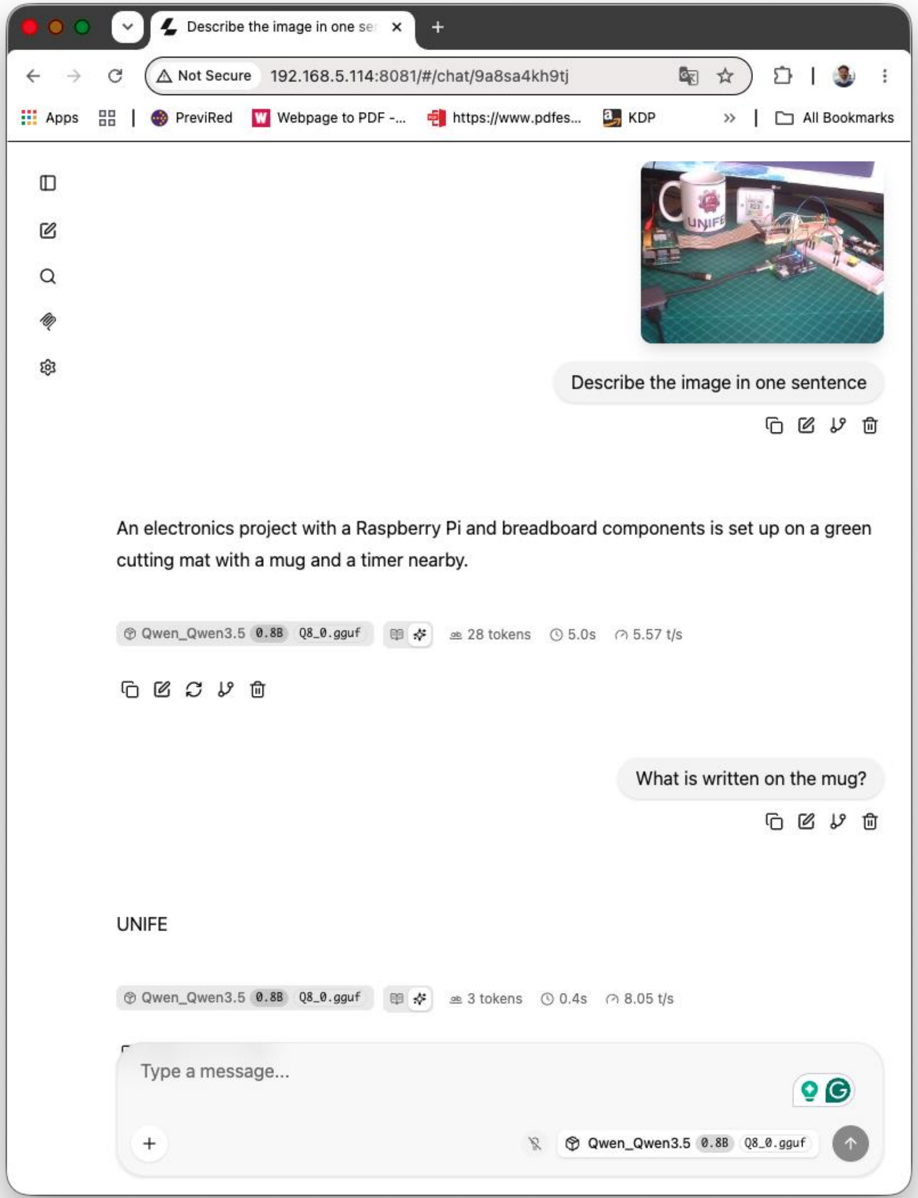
marcelo_rovai — mjrovai@raspi-5: ~ — ssh mjrovai@192.168.4.209 — 67×13

```
(ollama) mjrovai@raspi-5:~ $ ollama run gemma2:2b --verbose
>>> What is the capital of France?
The capital of France is **Paris**. 🗼


total duration:       4.373339337s
load duration:        48.129697ms
prompt eval count:    16 token(s)
prompt eval duration: 1.968114s
prompt eval rate:     8.13 tokens/s
eval count:           13 token(s)
eval duration:        2.313284s
eval rate:            5.62 tokens/s
```

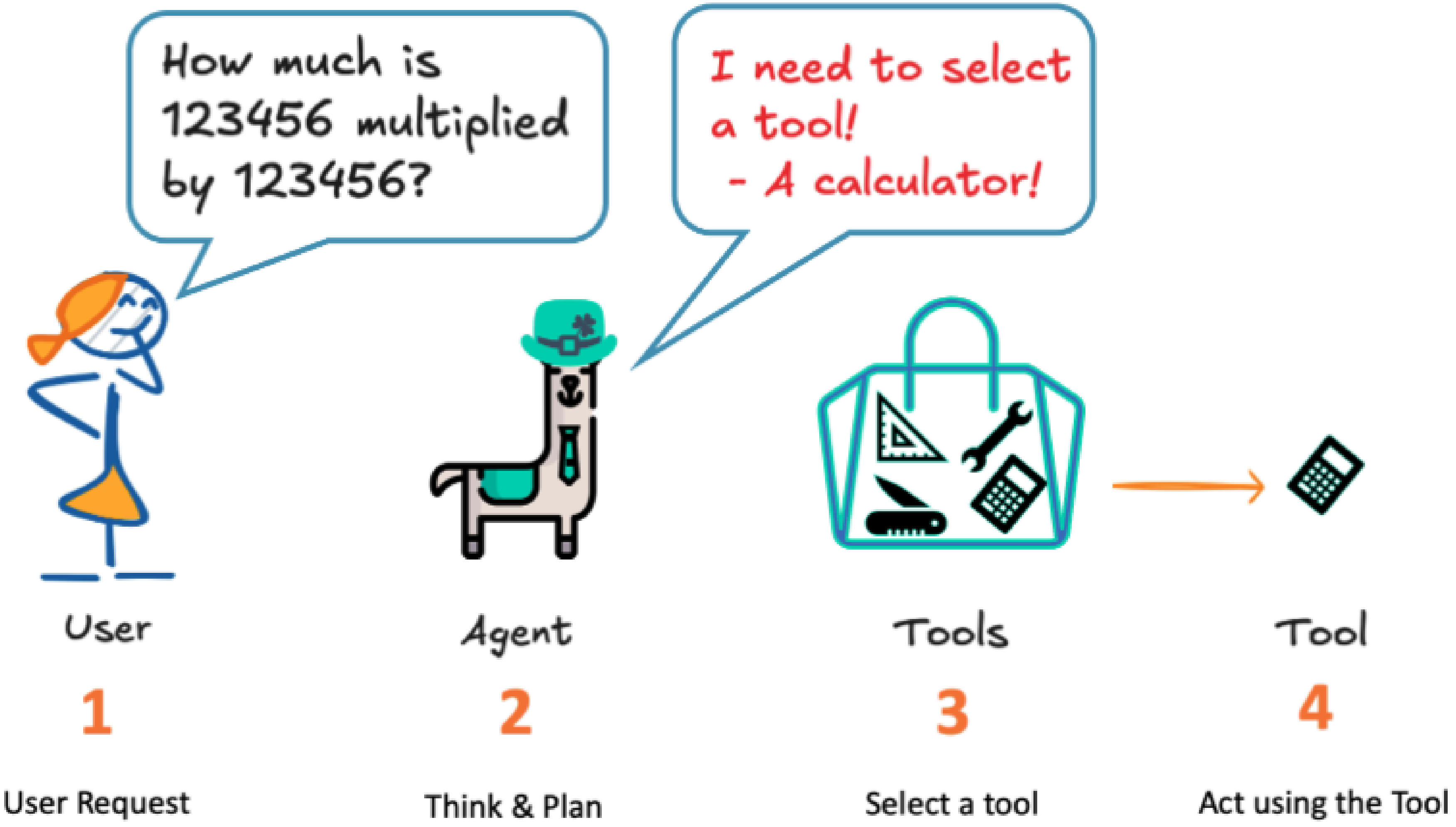
Small Models



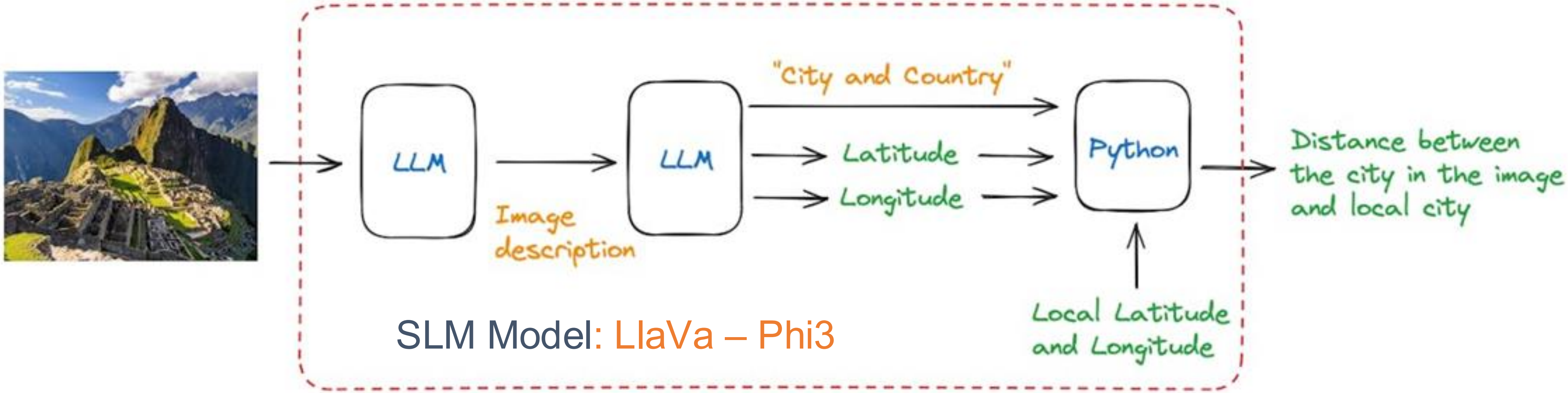
```
./build/bin/llama-server \
  -m        .../Qwen_Qwen3.5-0.8B-Q8_0.gguf \
  --mmproj .../mmproj-F16_0.8B.gguf \
  --host 0.0.0.0 --port 8081 \
  -c 4096 -t 4 -n 1024 \
  --jinja \
  --image-max-tokens 256 \
  --reasoning off --reasoning-budget 0
```

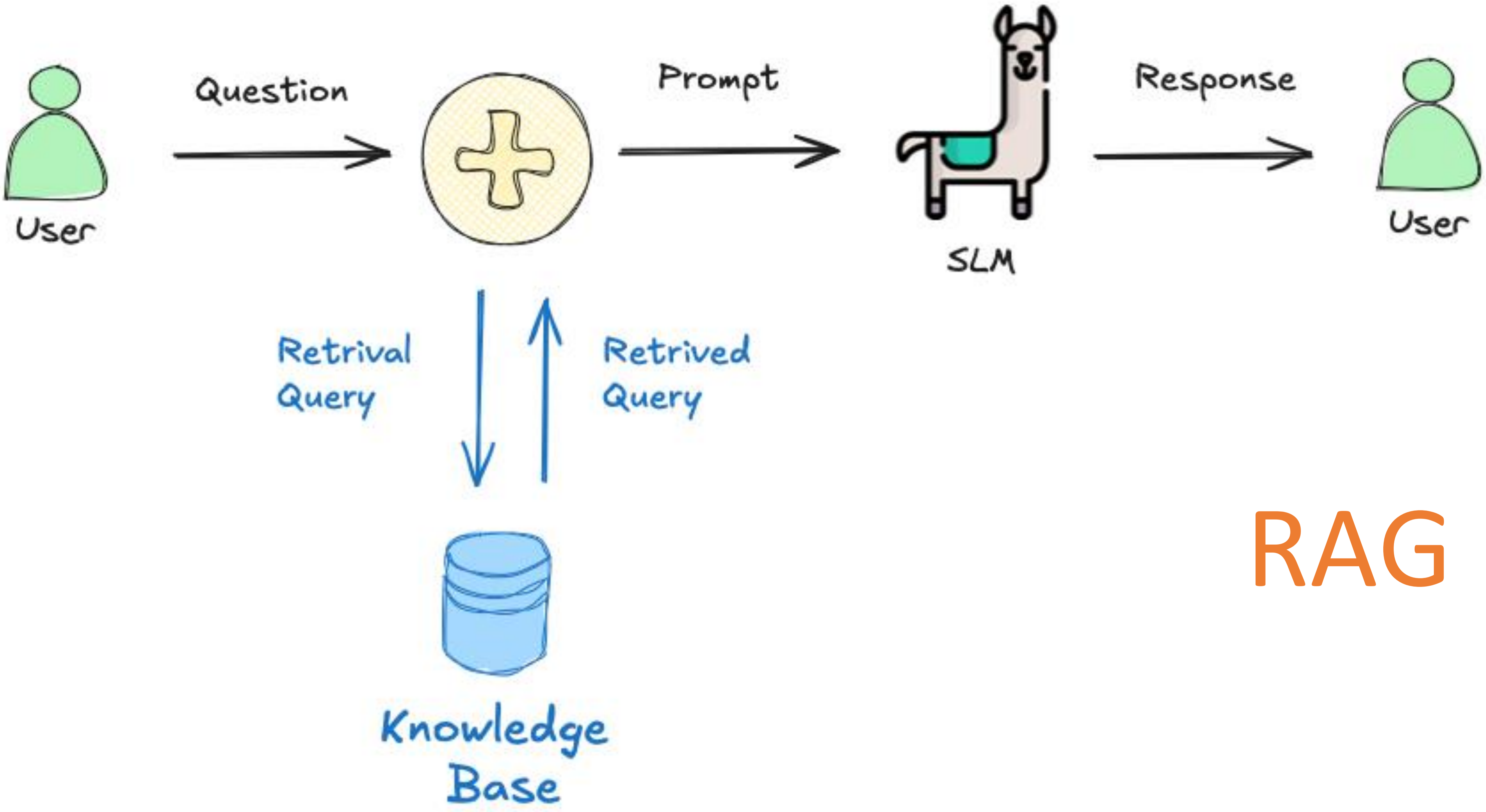


Agents


How much is 123456 multiplied by 123456?
I need to select a tool!
- A calculator!
User
1
User Request
Agent
2
Think & Plan
Tools
3
Select a tool
Tool
4
Act using the Tool

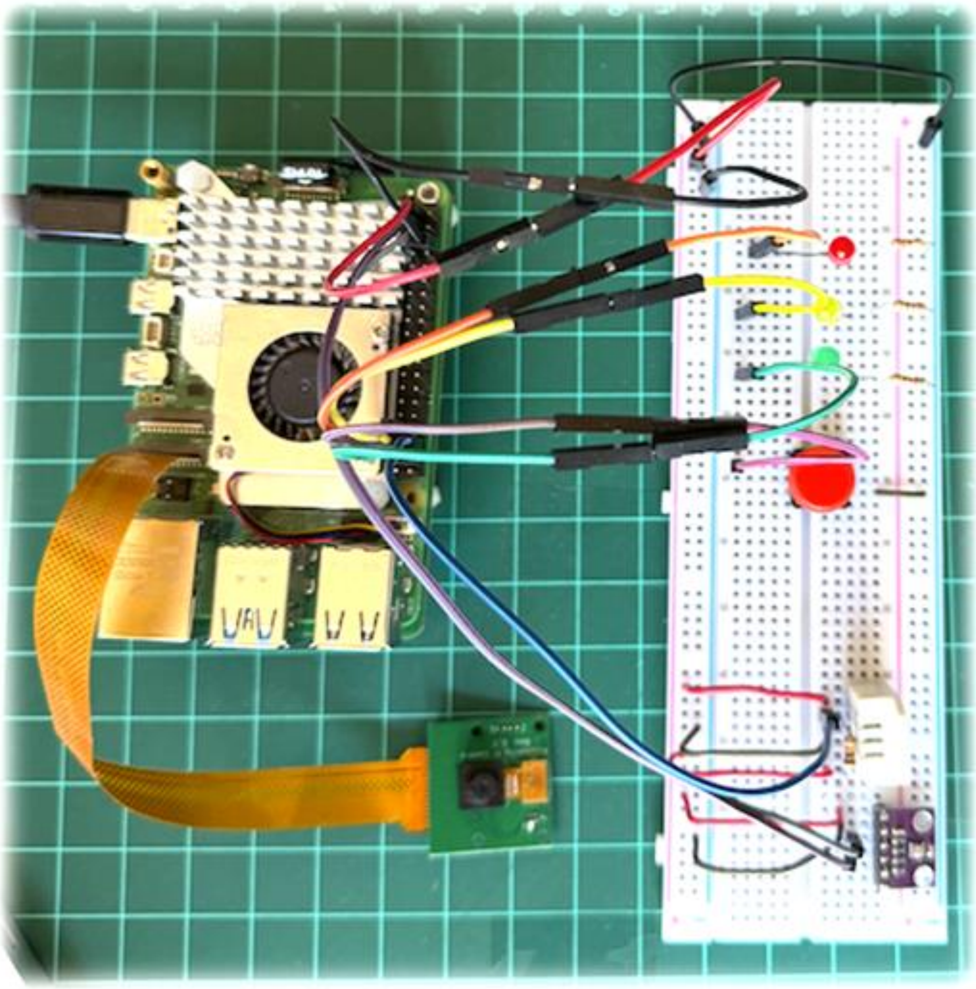
Function Calling

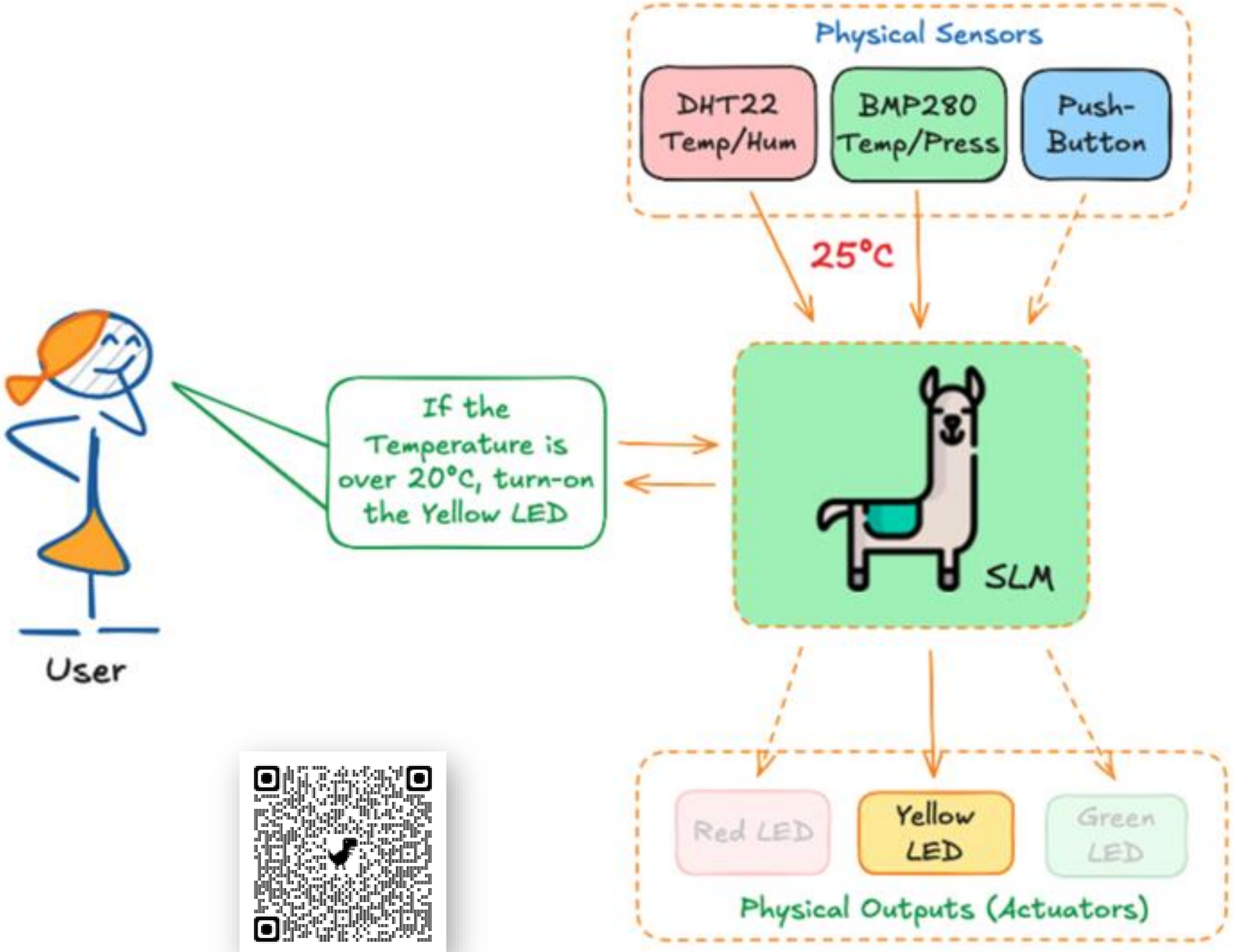



Question
Prompt
Response
User
SLM
User
Retrival
Query
Retrived
Query
Knowledge
Base

RAG

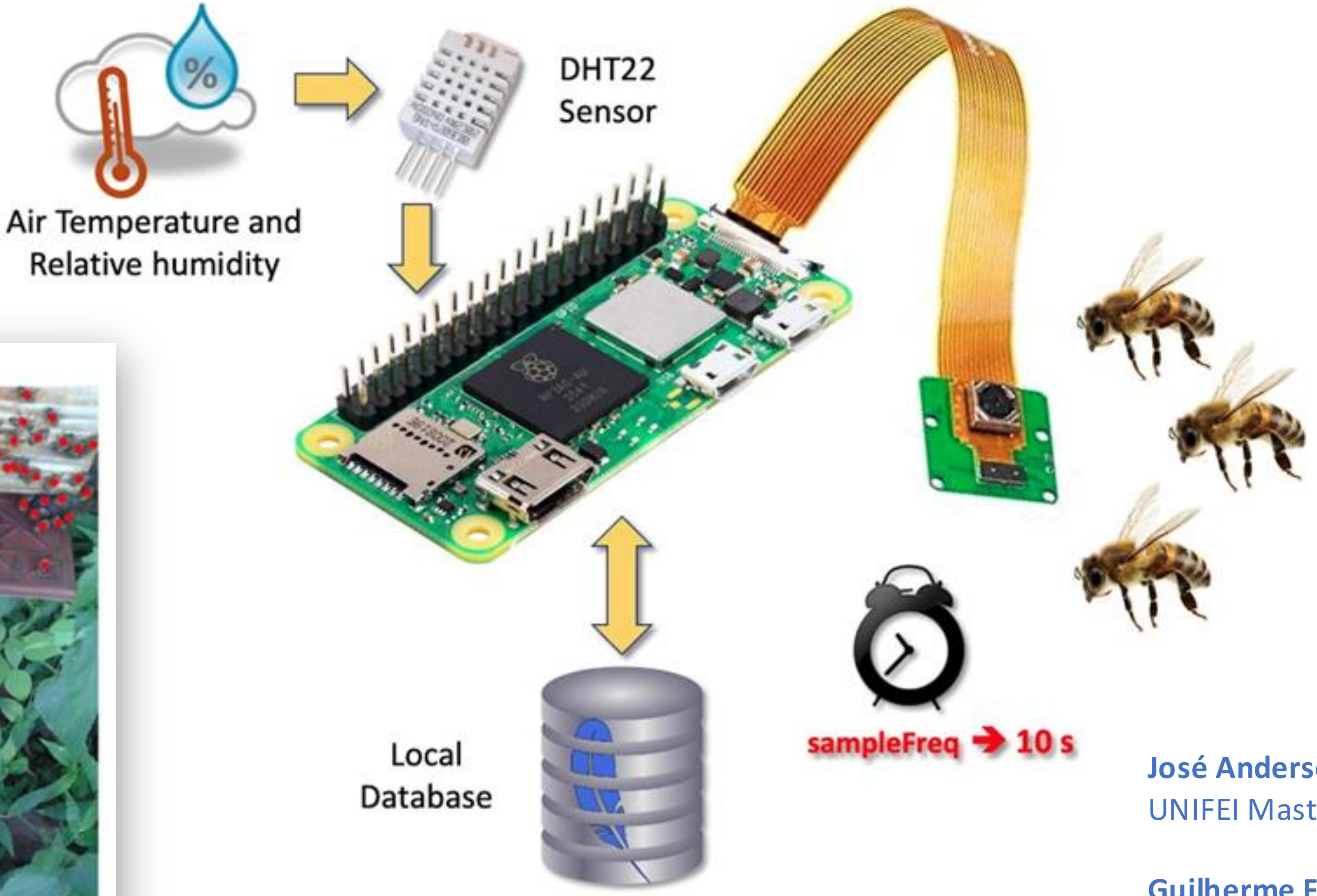
Physical Computing Meets Intelligence





Real-World Applications

Bee Counting



José Anderson Reis
UNIFEI Master's Student

Guilherme Fernandes
UNIFEI Grad Student

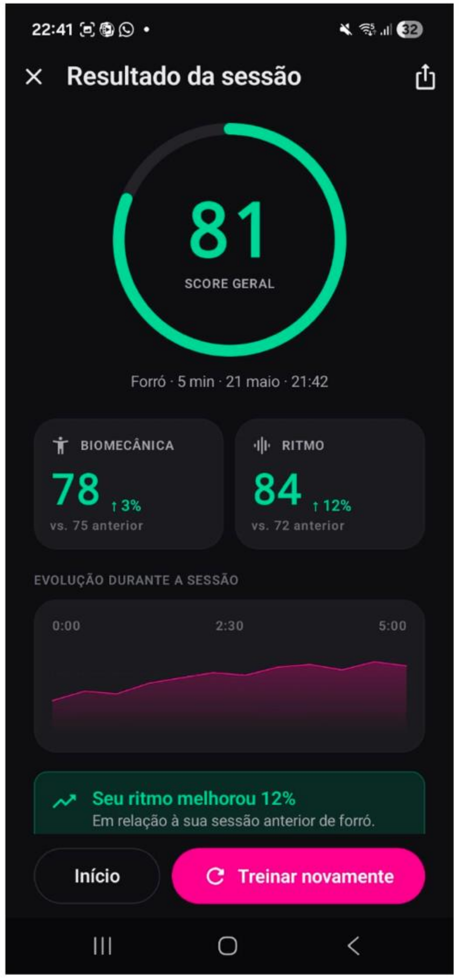
Ant Detection

FURG UTEC UNRaf

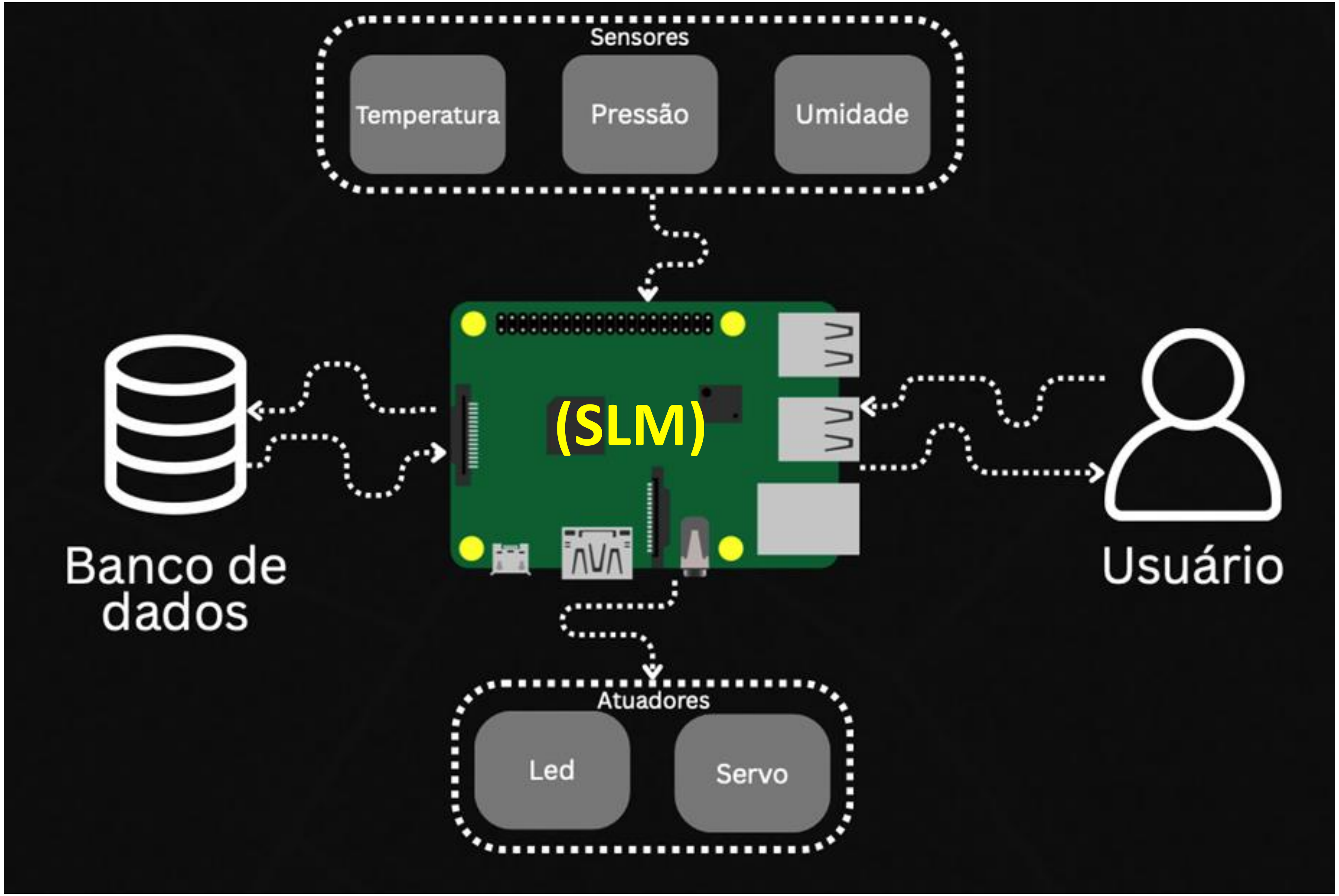
Juan Camilo Abedala
PRIA

Dança AI

Sistema Embarcado de Visão Computacional para Análise Biomecânica e Cadência Rítmica em Dança de Salão



Felipe Santana Medeiros
João Luiz de Miranda Cilli
UNIFEI

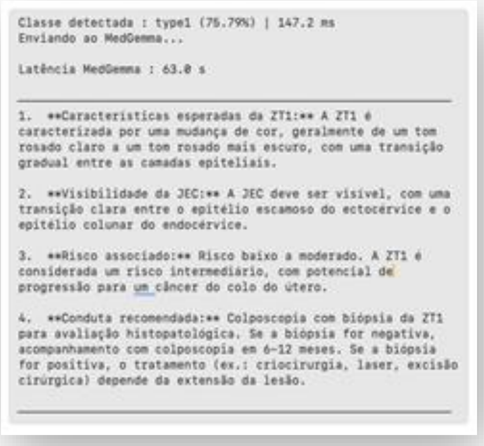


João Pedro Fonseca
UNIFEI Grad Student (IESTI)

Matheus Souza
UNIFEI Grad Student (IMC)

Intelligent Colposcope Image Classification System

BRUNA CUSTÓDIO ALVES,
VIVIAN LEITE FRAGOSO
UNIFEI



YOLO 26 → MobiileNet V3 → MedGemma

About the Course
& Syllabus

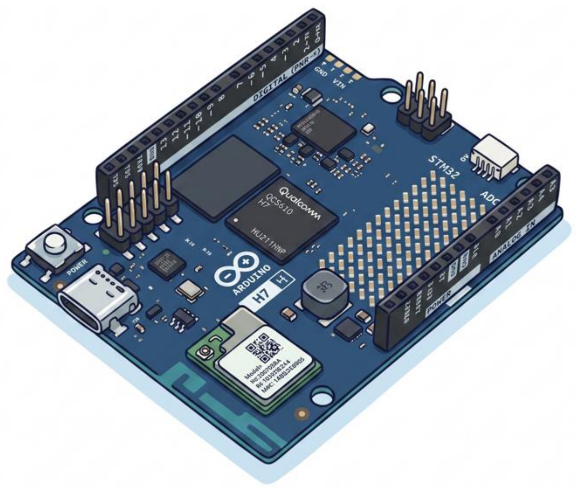
Hands-on Learning

- **Software / Tools**
  - Linux
  - Python
  - TensorFlow and PyTorch
  - Google Colab or Jupyter Notebook
  - Edge Impulse Studio
  - Ultralytics and RoboFlow
  - LangChain
  - Ollama
  - Llama.cpp
- **Hardware**
  - Raspberry Pi Zero W2
  - Arduino UNO Q
  - Raspberry Pi 5
  - Miscellaneous items such as LEDs, Sensors, etc.

Part 1: Fixed Function AI (Reactive)

Part 2: Generative AI (Proactive)

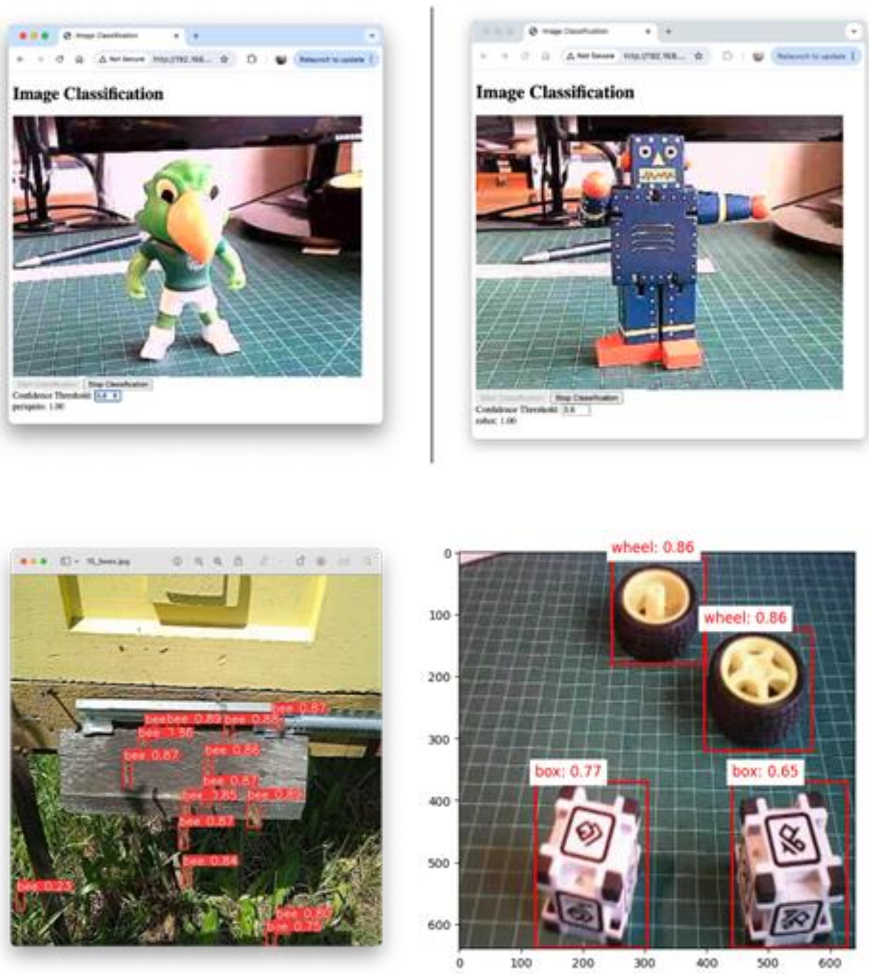
Raspberry Pi Zero W2



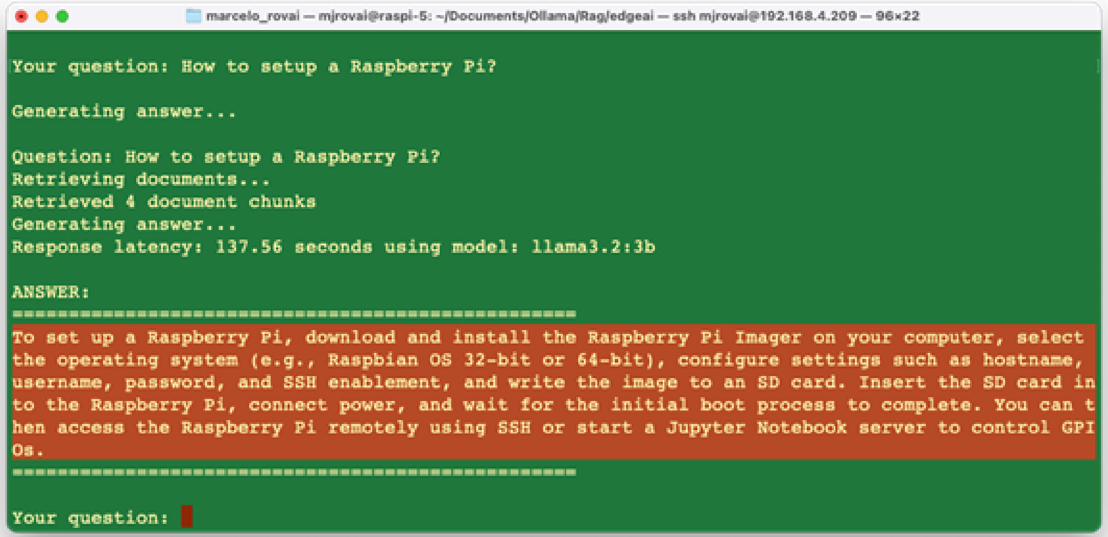
Arduino UNO Q

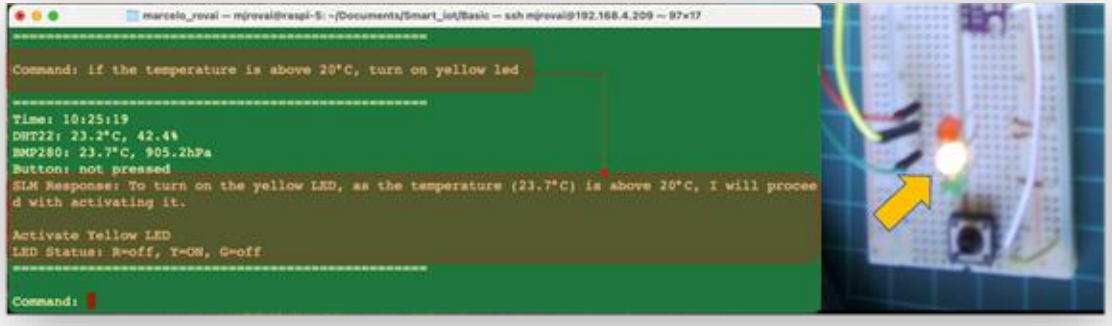
Raspberry Pi 5

Part 1: Fixed Function AI (Reactive)

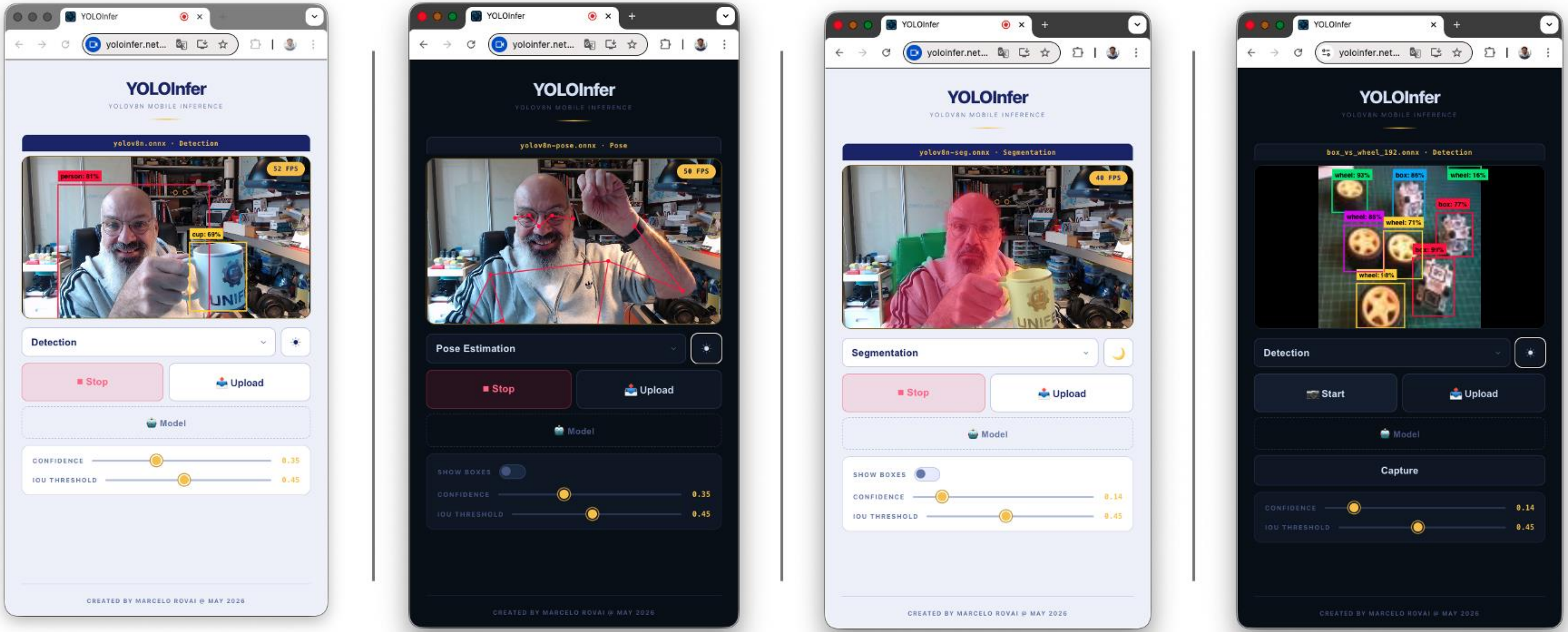


Part 2: Generative AI (Proactive)





YOLOInfer - Mobile YOLO Inference



Course Structure

The course is divided into two main parts (based on the textbook: Edge AI Engineering by Marcelo Rovai):

- **Part 1 (Weeks 1-7): Fixed Function AI** - Focus on image classification and object detection
- **Part 2 (Weeks 8-15): Generative AI** - Focus on Small Language Models (SLMs)

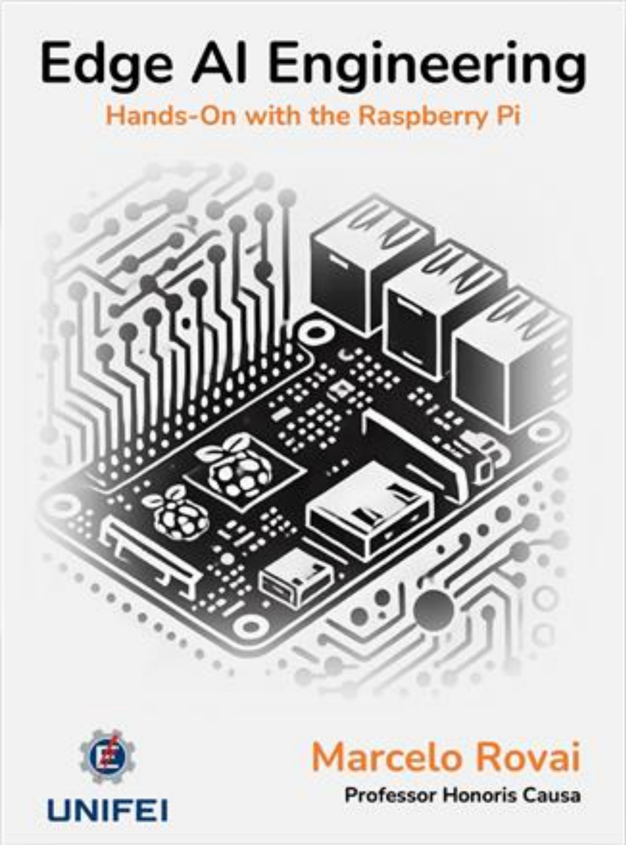
Key Learning Areas

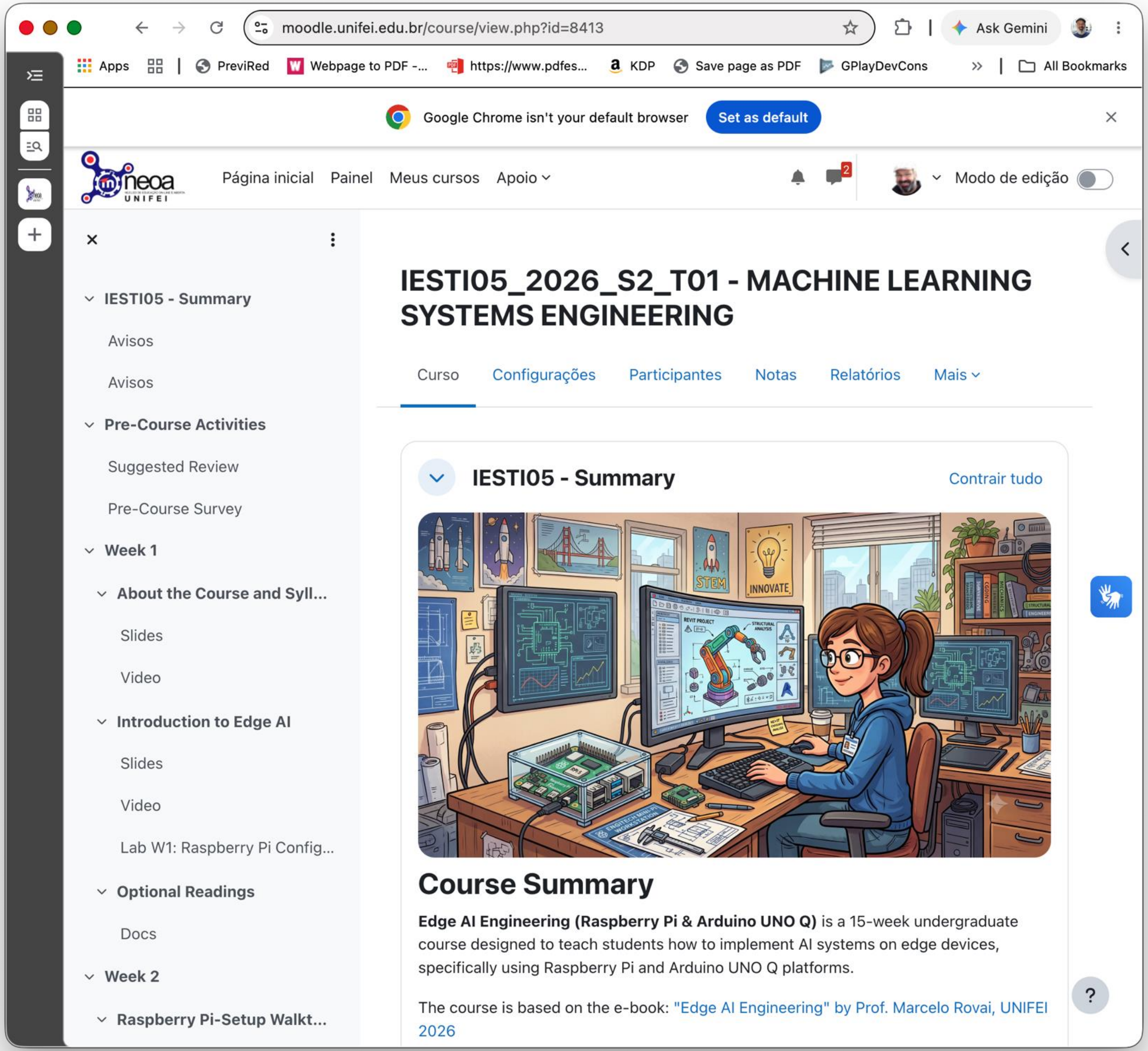
- Raspberry Pi and Arduino UNO Q setup, configuration, and optimization
- Computer vision and TensorFlow Lite
- Image classification and object detection implementation
- Small Language Model deployment and integration
- Retrieval-Augmented Generation (RAG) systems and Agents
- Physical computing integration with sensors and actuators

Assessment Structure

- **40%** - Weekly hands-on labs, quizzes, and surveys
- **20%** - Midterm project (Fixed Function AI system)
- **30%** - Final project (Generative AI application)
- **10%** - Participation and documentation

EdgeAI Engineering




moodle.unifei.edu.br/course/view.php?id=8413
Ask Gemini
Apps
PreviRed
Webpage to PDF -...
https://www.pdfes...
KDP
Save page as PDF
GPlayDevCons
All Bookmarks
Google Chrome isn't your default browser
Set as default
neoa
UNIFEI
Página inicial
Painel
Meus cursos
Apoio
Modo de edição
IESTI05 - Summary
Avisos
Avisos
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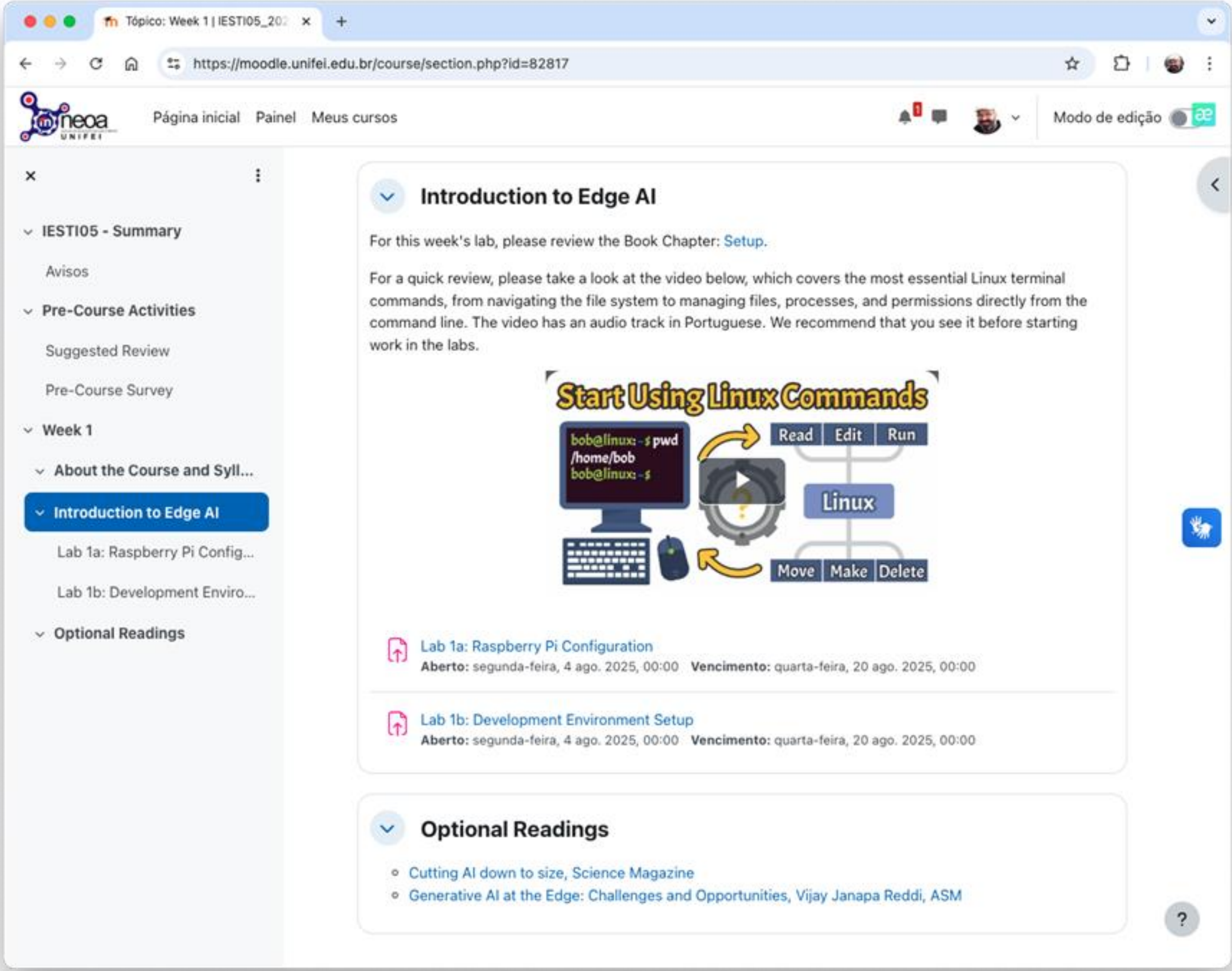
Course Summary
Edge AI Engineering (Raspberry Pi & Arduino UNO Q) is a 15-week undergraduate course designed to teach students how to implement AI systems on edge devices, specifically using Raspberry Pi and Arduino UNO Q platforms.
The course is based on the e-book: "Edge AI Engineering" by Prof. Marcelo Rovai, UNIFEI 2026


Tópico: Week 1 | IESTI05_202
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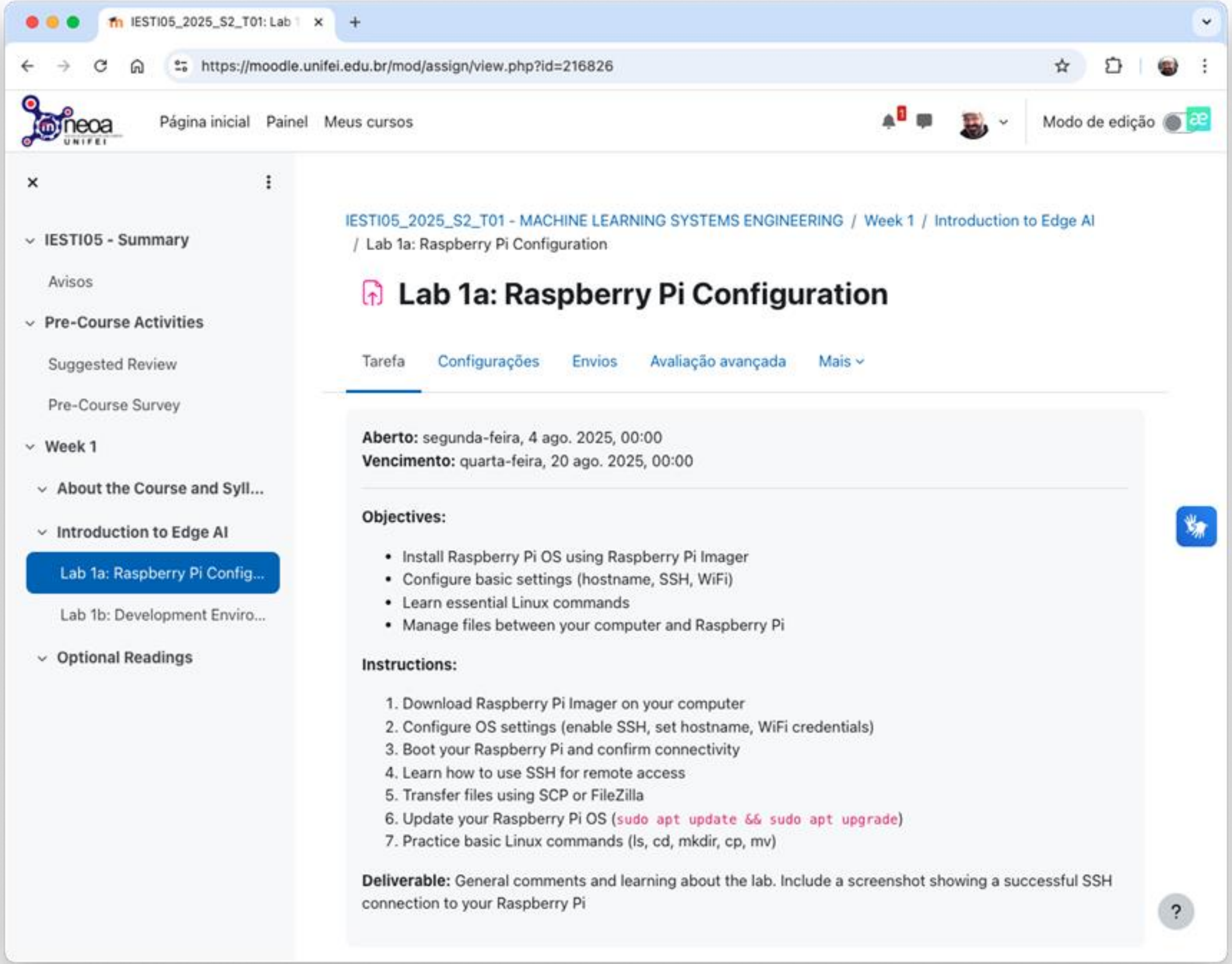
Introduction to Edge AI
For this week's lab, please review the Book Chapter: Setup.
For a quick review, please take a look at the video below, which covers the most essential Linux terminal commands, from navigating the file system to managing files, processes, and permissions directly from the command line. The video has an audio track in Portuguese. We recommend that you see it before starting work in the labs.
Start Using Linux Commands
bob@linux:~$ pwd
/home/bob
bob@linux:~$
Read
Edit
Run
Linux
Move
Make
Delete
Lab 1a: Raspberry Pi Configuration
Aberto: segunda-feira, 4 ago. 2025, 00:00 Vencimento: quarta-feira, 20 ago. 2025, 00:00
Lab 1b: Development Environment Setup
Aberto: segunda-feira, 4 ago. 2025, 00:00 Vencimento: quarta-feira, 20 ago. 2025, 00:00
Optional Readings
Cutting AI down to size, Science Magazine
Generative AI at the Edge: Challenges and Opportunities, Vijay Janapa Reddi, ASM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Lab 1a: Raspberry Pi Configuration
Tarefa
Configurações
Envios
Avaliação avançada
Mais
Aberto: segunda-feira, 4 ago. 2025, 00:00
Vencimento: quarta-feira, 20 ago. 2025, 00:00
Objectives:
• Install Raspberry Pi OS using Raspberry Pi Imager
• Configure basic settings (hostname, SSH, WiFi)
• Learn essential Linux commands
• Manage files between your computer and Raspberry Pi
Instructions:
1. Download Raspberry Pi Imager on your computer
2. Configure OS settings (enable SSH, set hostname, WiFi credentials)
3. Boot your Raspberry Pi and confirm connectivity
4. Learn how to use SSH for remote access
5. Transfer files using SCP or FileZilla
6. Update your Raspberry Pi OS (sudo apt update && sudo apt upgrade)
7. Practice basic Linux commands (ls, cd, mkdir, cp, mv)
Deliverable: General comments and learning about the lab. Include a screenshot showing a successful SSH connection to your Raspberry Pi
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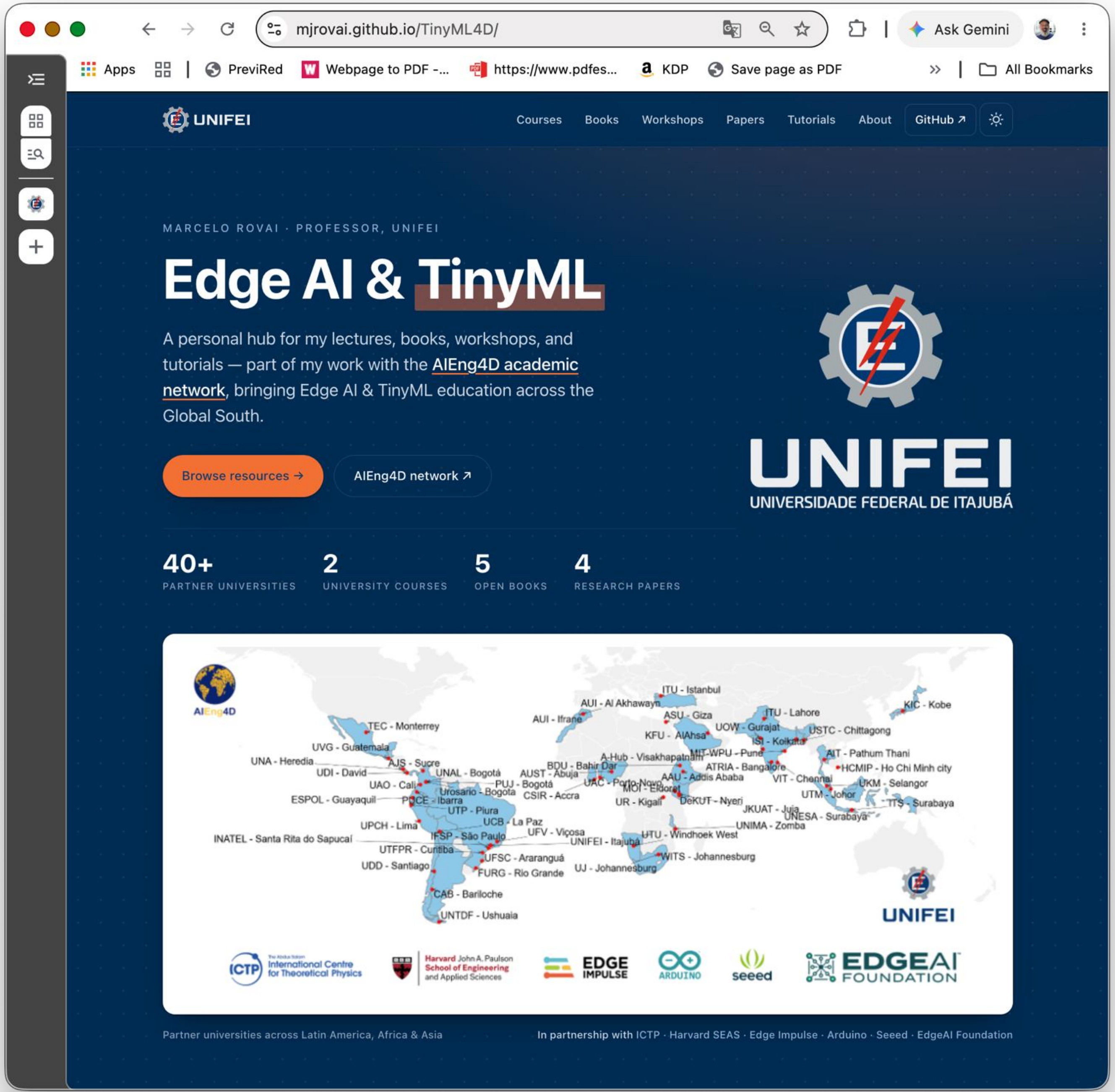
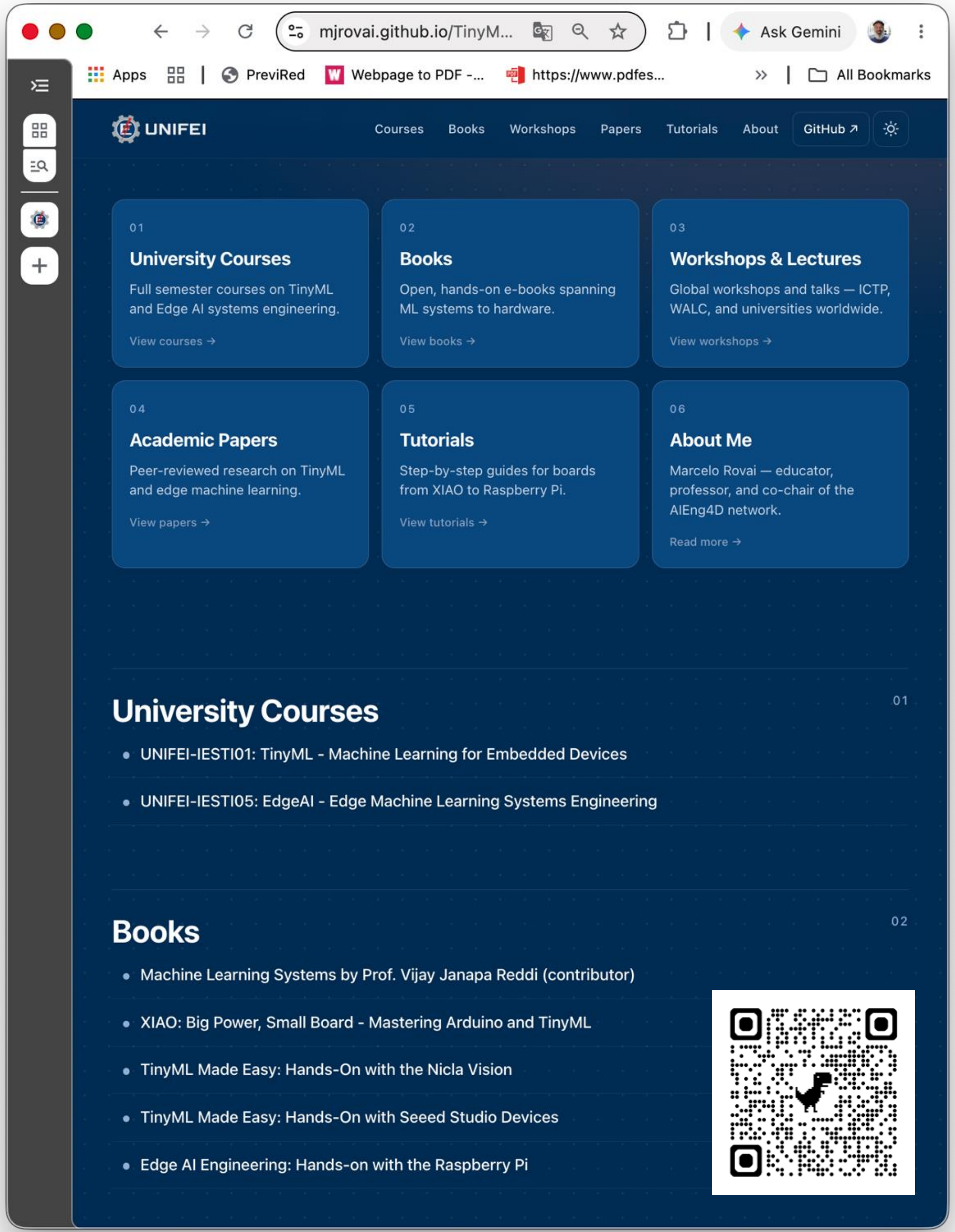
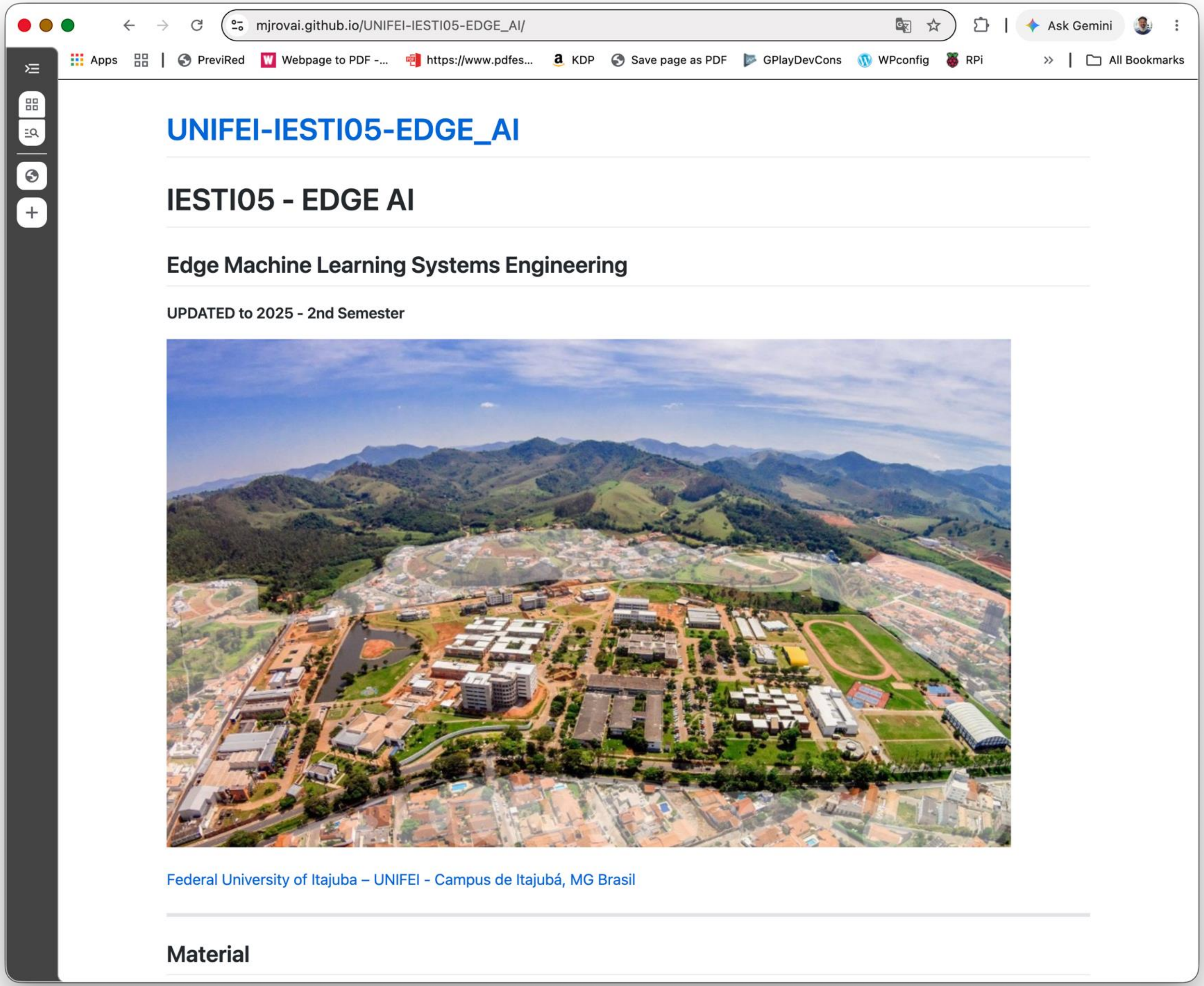


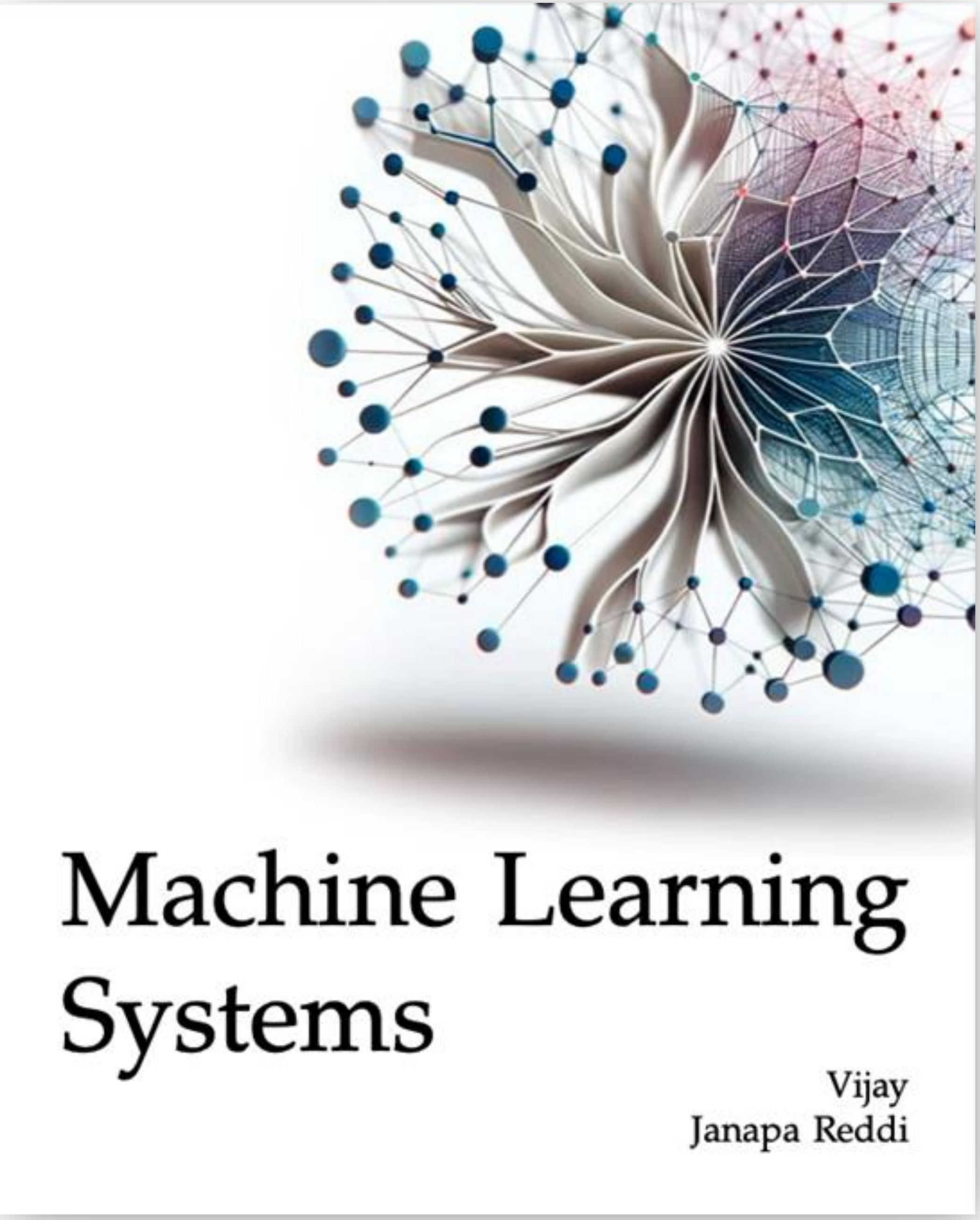


https://mjrovai.github.io/TinyML4D/



https://mjrovai.github.io/UNIFEI-IESTI05-EDGE_AI/

Learning Deeper



https://mlsysbook.ai/

Questions?

Prof. Marcelo J. Rovai

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